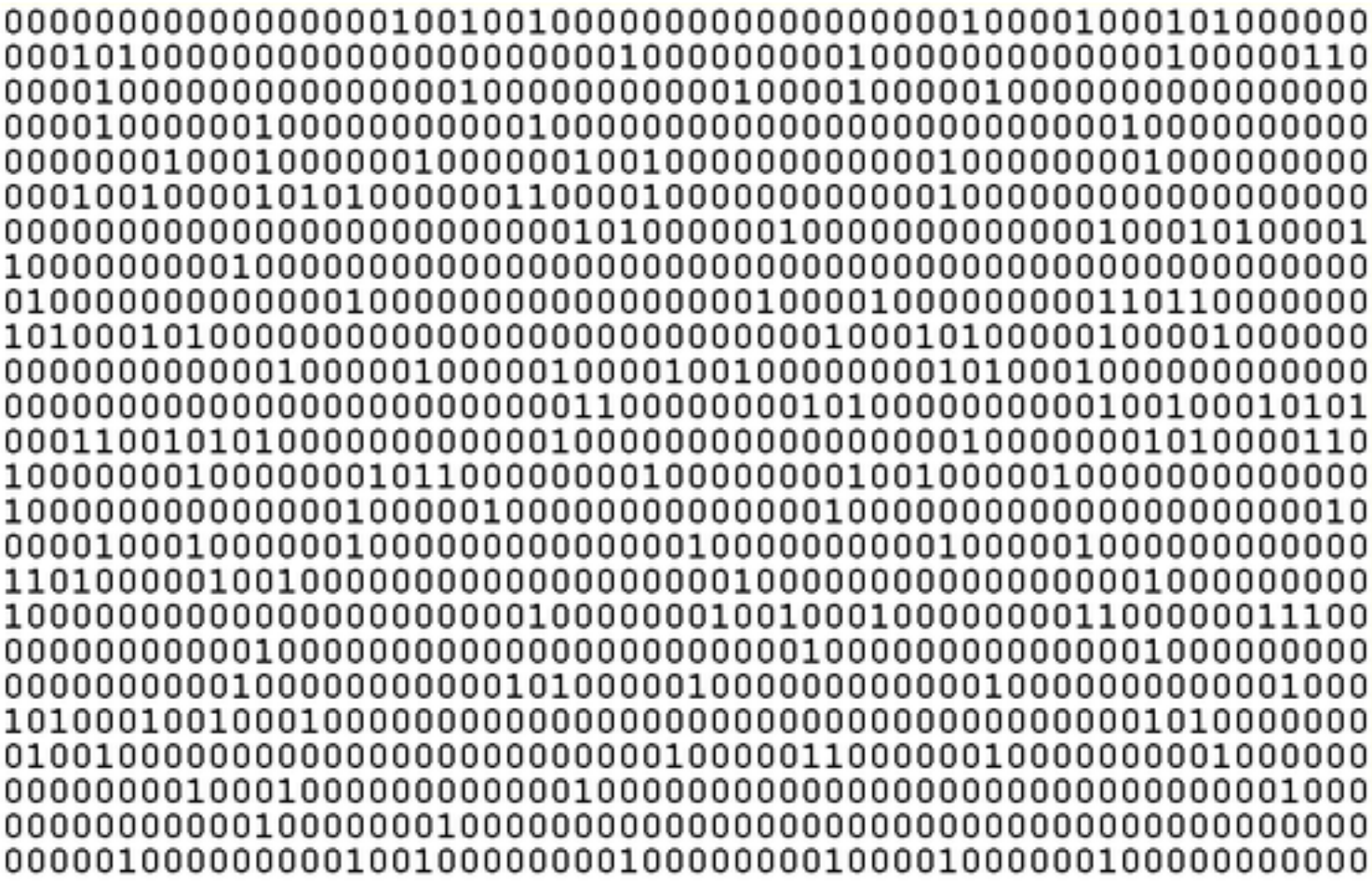
Lecture 3

- Evidence for

$$h(x) = \log_2 \frac{1}{P(x)}$$

- Source coding theorem

How to compress a redundant file?



e.g., $N = 1000$ tosses of a bent coin with $p_1 = 0.1$

How to measure information content?

Claims:

1. The *Shannon information content* of an outcome

$$h(x=a_i) = \log_2 \frac{1}{P(x=a_i)}$$

is a sensible measure of information content.

2. The entropy

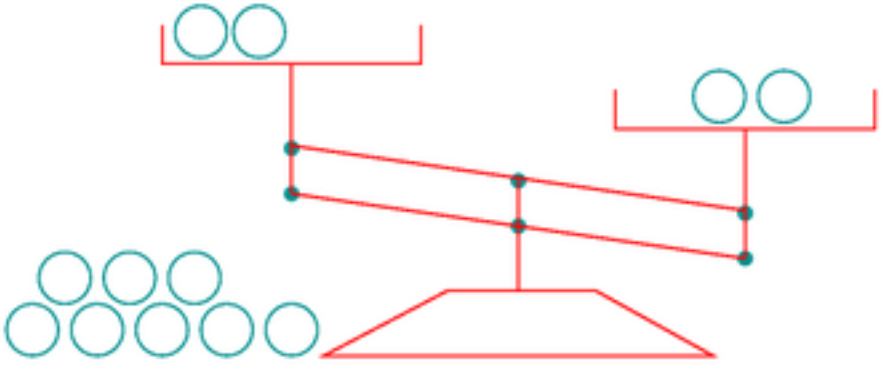
$$H(X) = \sum_x P(x) \log_2 \frac{1}{P(x)}$$

is a sensible measure of expected information content.

(sketch h(p))
Point out additivity of h.

Testing 'Shannon information content' claims

- Weighing problem



- Shannon says

  Most 'informative' experiment is the one with maximum entropy

How to measure information content?

Claims:

1. The *Shannon information content* of an outcome

$$h(x=a_i) = \log_2 \frac{1}{P(x=a_i)}$$

is a sensible measure of information content.

2. The entropy

$$H(X) = \sum_x P(x) \log_2 \frac{1}{P(x)}$$

is a sensible measure of expected information content.

3. Source coding theorem –
N outcomes from a source X can be compressed into roughly $NH(X)$ bits.

The Bent Coin Lottery

A coin with $p_1 = 0.1$ will be tossed $N = 1000$ times.

The outcome is $\mathbf{x} = x_1 x_2 \ldots x_N$.

e.g., $\mathbf{x} =$ 000001001000100...00010

You can buy any of the 2^N possible tickets for £1 each, before the coin-tossing.

If you own ticket $\mathbf{x}$, you win £1,000,000,000.

Q To have a 99% chance of winning, at lowest possible cost, which tickets would you buy?

- And how many tickets is that?

  Express your answer in the form $2^{(\cdots)}$.

Lottery tickets available

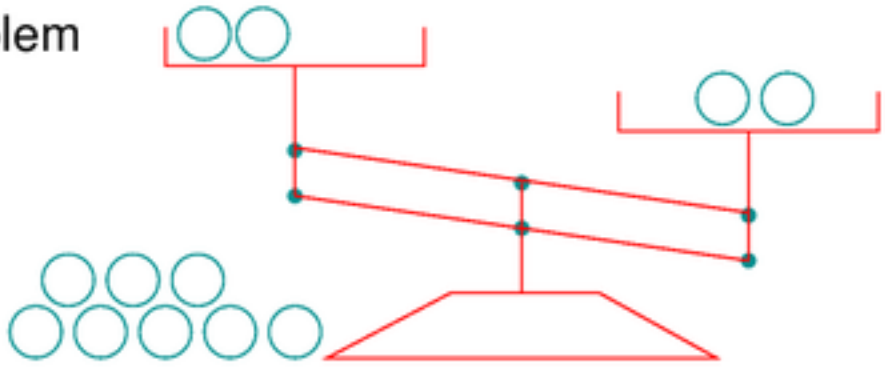
2^N

0000000000.....00000
0000000000.....00001
0000000000.....00010
0000000000.....00011
0000000000.....00100
0000000000.....00101
0000000000.....00110
0000000000.....00111
⋮
0010000001.....01000
⋮
1111111111.....11101
1111111111.....11110
1111111111.....11111

We'll come back to this in a moment. First...

Testing 'Shannon information content' claims

- Weighing problem

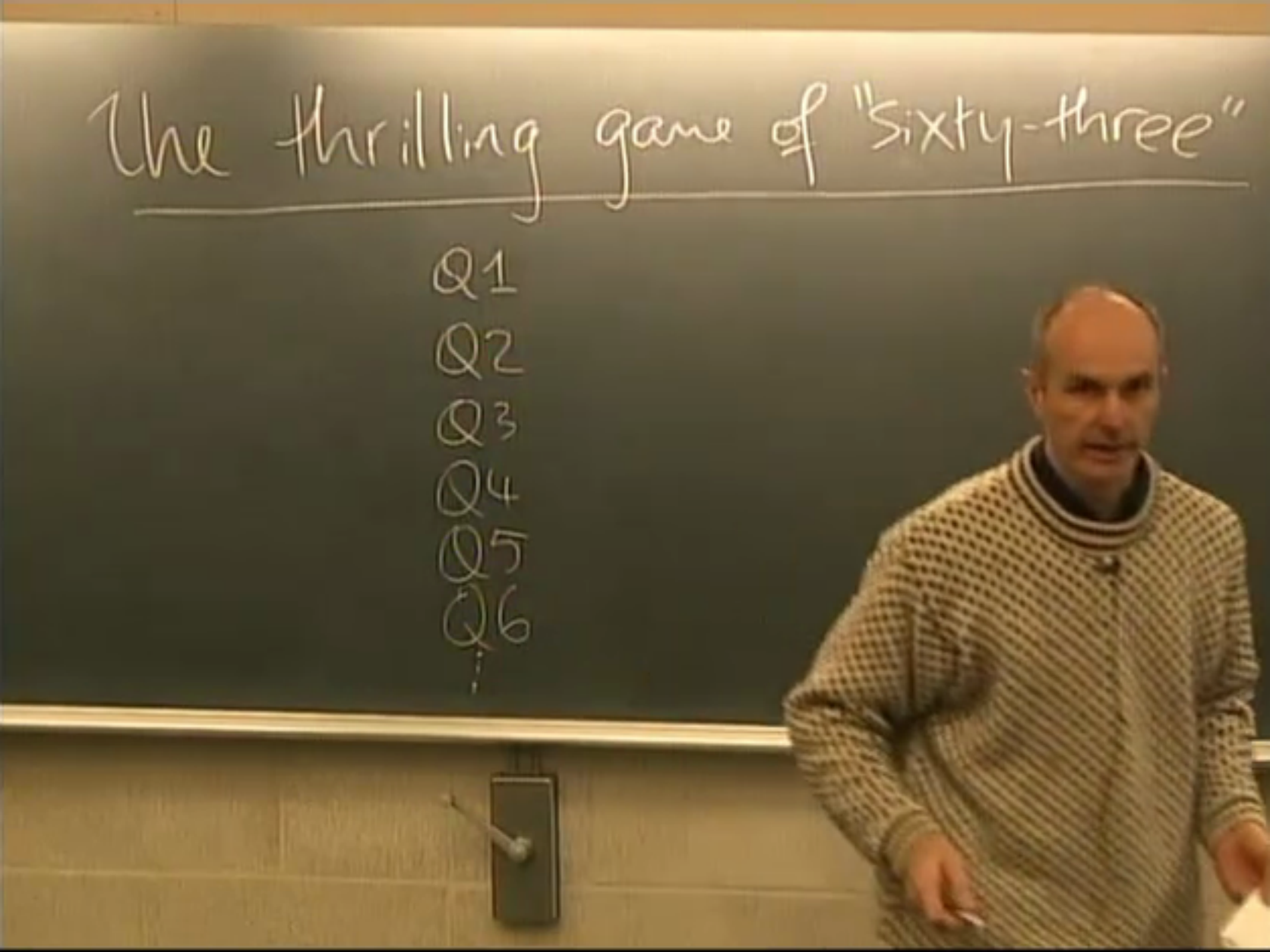


- Sixty-three

- Submarine

Sixty-three

$$x \in \{0, 1, 2, 3, 4, 5, 6, \ldots, 63\}$$


The thrilling game of "Sixty-three"
Q1
Q2
Q3
Q4
Q5
Q6
⋮

of "sixty-three"

$x \geq 32$? 1 Yes

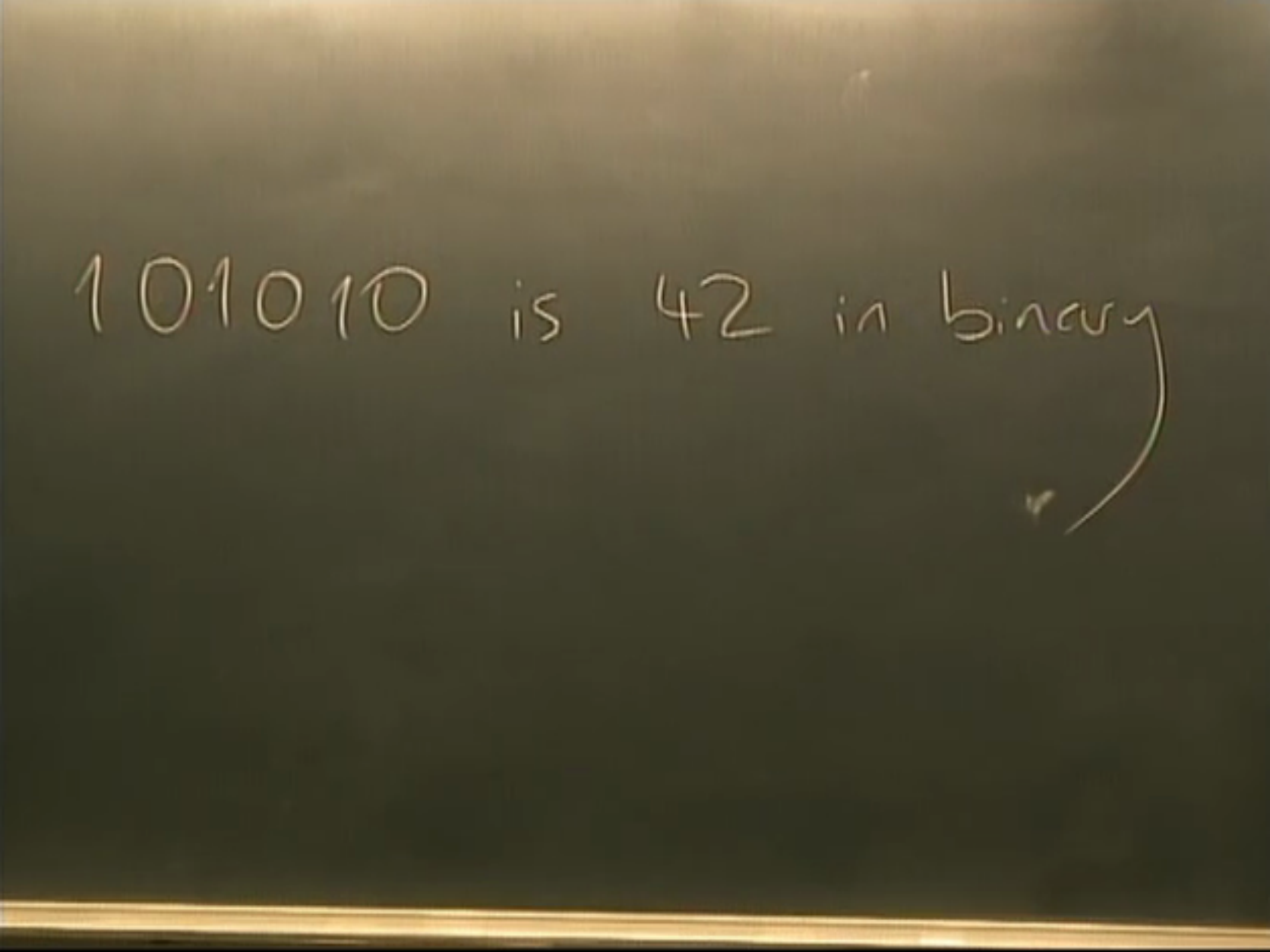
$x \bmod 32 \geq 16$? 48, ... 63 0

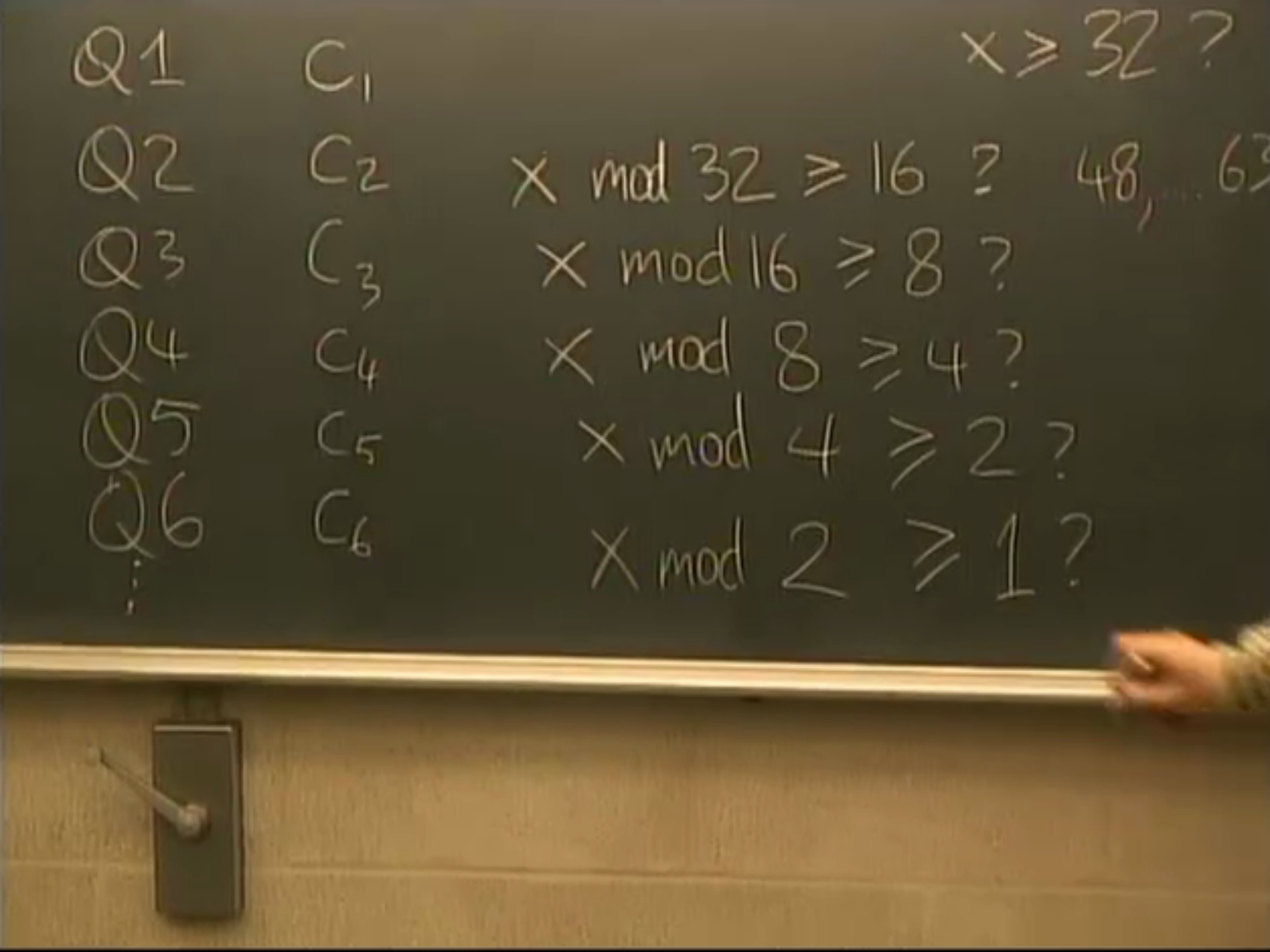
$x \bmod 16 \geq 8$? 1

$x \bmod 8 \geq 4$? 0

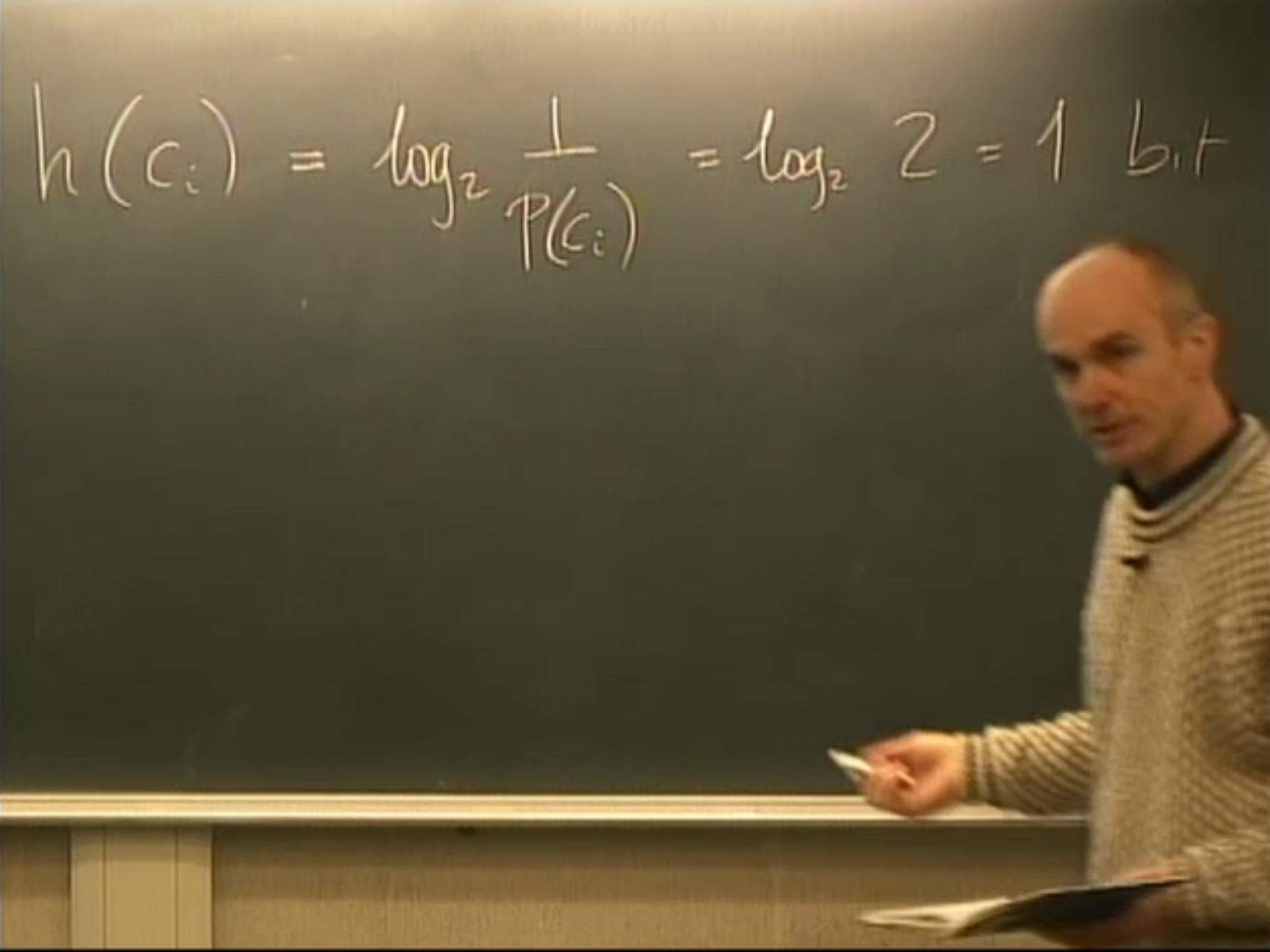
$x \bmod 4 \geq 2$? 1

$x \bmod 2 \geq 1$? 0


101010 is 42 in binary


Q1 C1 x ≥ 32 ?
Q2 C2 x mod 32 ≥ 16 ? 48, ... 63
Q3 C3 x mod 16 ≥ 8 ?
Q4 C4 x mod 8 ≥ 4 ?
Q5 C5 x mod 4 ≥ 2 ?
Q6 C6 x mod 2 ≥ 1 ?

$$h(c_i) = \log_2 \frac{1}{P(c_i)} = \log_2 2 = 1 \text{ bit}$$



$$h(c_i) = \log_2 \frac{1}{P(c_i)} = \log_2 2 = 1 \text{ bit}$$

Total Shannon Info content gained = 6

The string $c_1 \ldots c_6$

$\frac{1}{P(c_i)} = \log_2 2 = 1$ bit

content gained = 6 bits

$c_1 \ldots c_6$ is an encoding of x

$C(x)$ $C(42) = 101010$

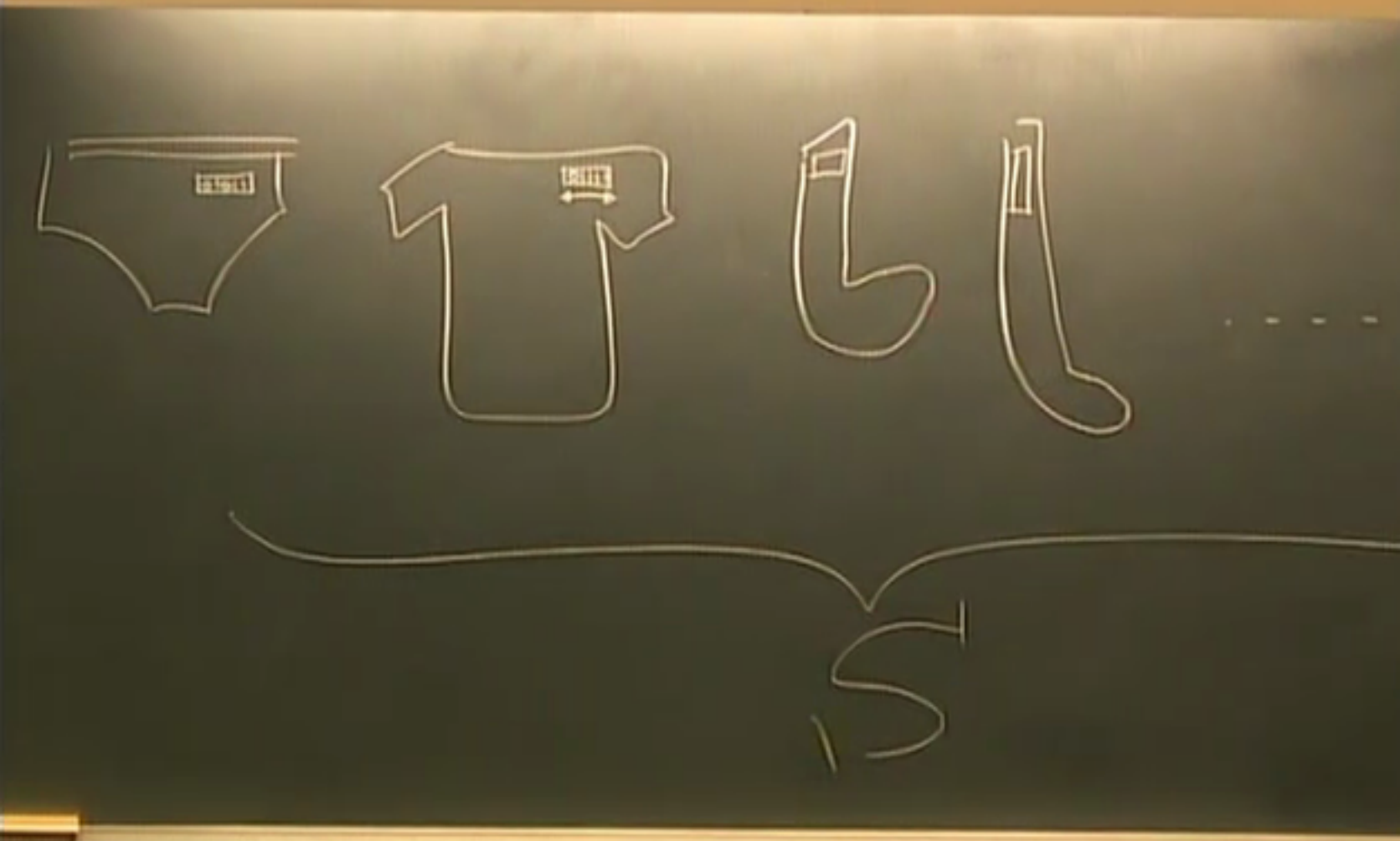
Sixty-three

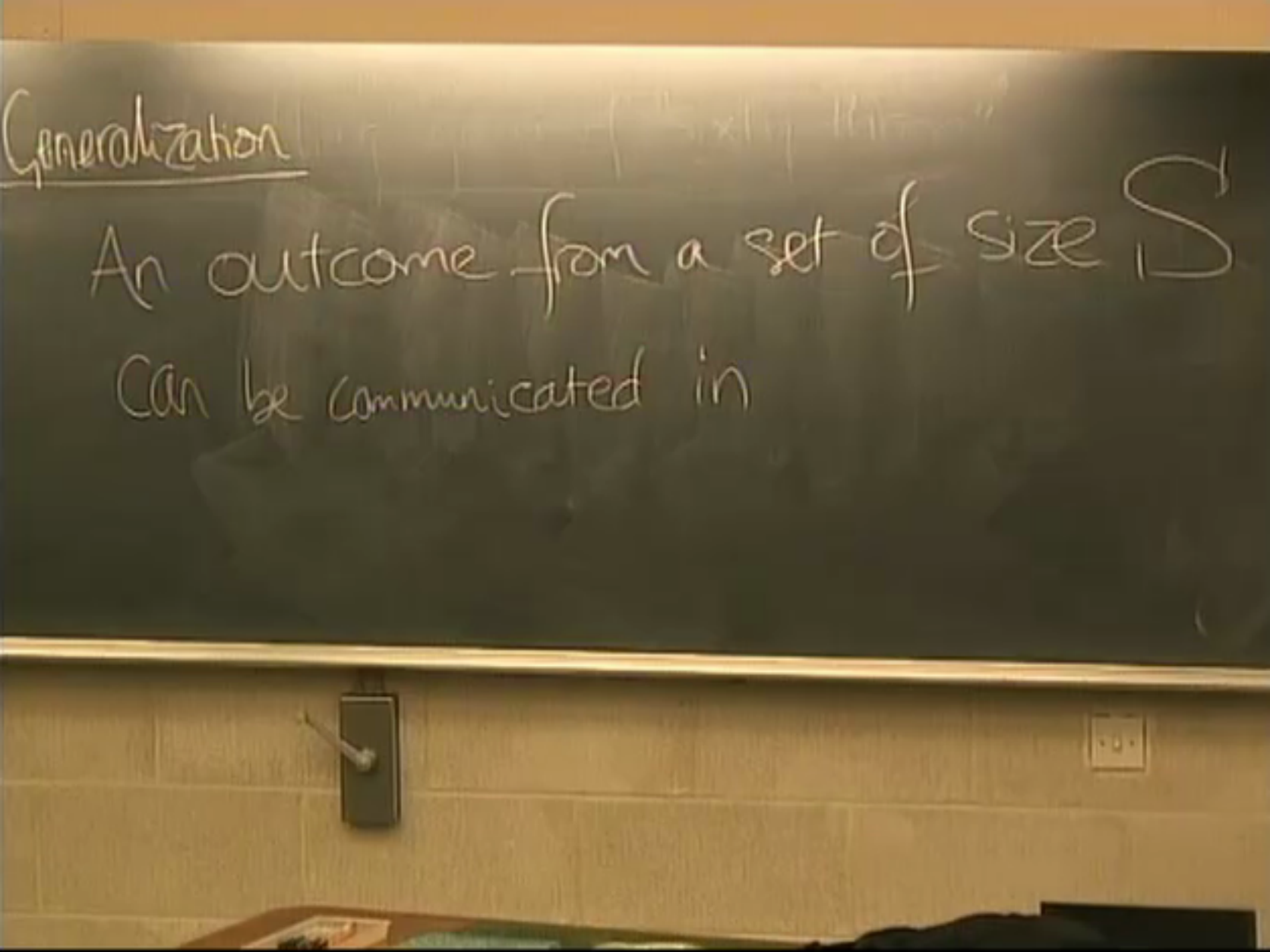
We were able to give unique binary names to every one of 64 outcomes if each name was 6 bits long.

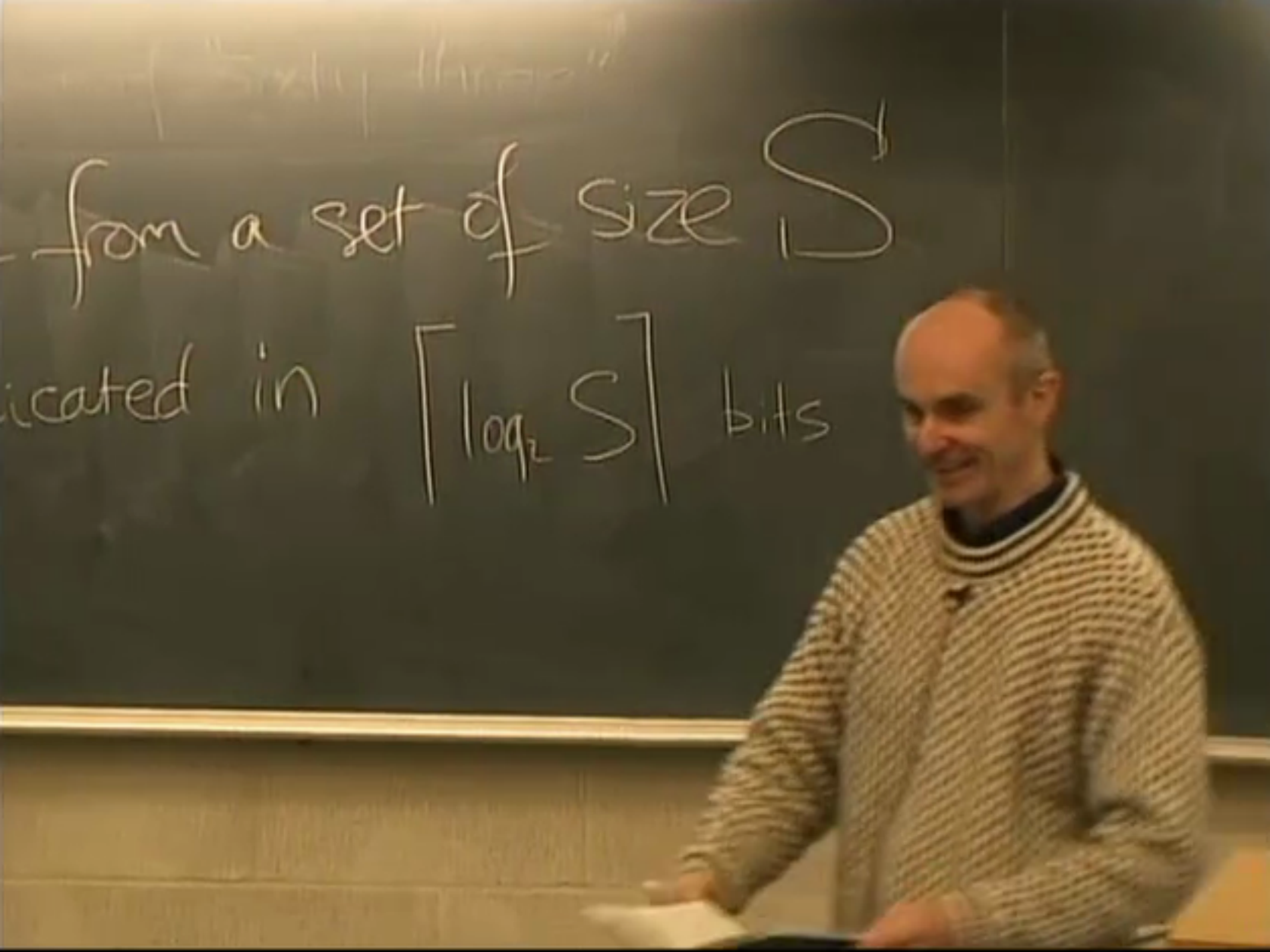
e.g., $c(42) = 101010$
$c(20) = 010100$

Generalize –

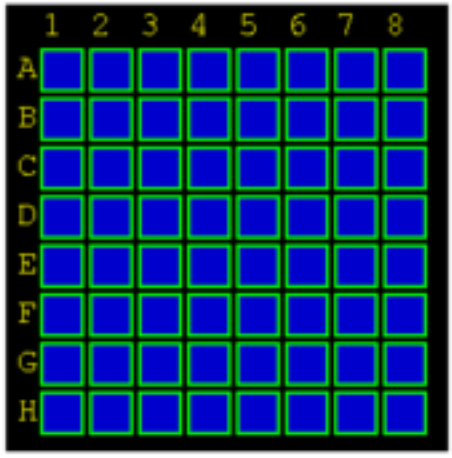
Q If there are S possible outcomes, how many bits long must each name be, if each outcome has a unique name?




Generalization
An outcome from a set of size S
can be communicated in


from a set of size S
icated in $\lceil \log_2 S \rceil$ bits

Submarine



Submarine

Restart

Submarine

Dismiss

Unhit subs: 1

Unhit squares: 63

Probabilities:

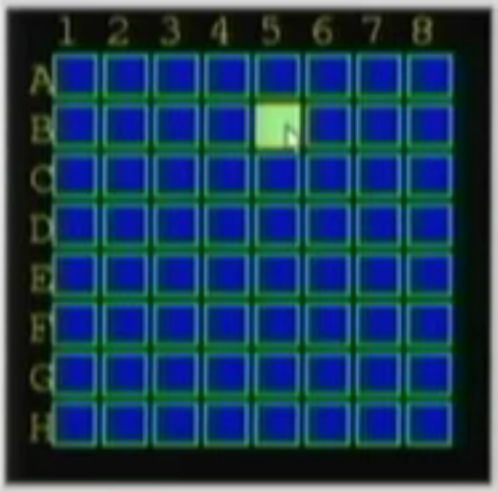
P(y): 0/1 0.0

P(n): 1/1 1.0

Entropy: 0

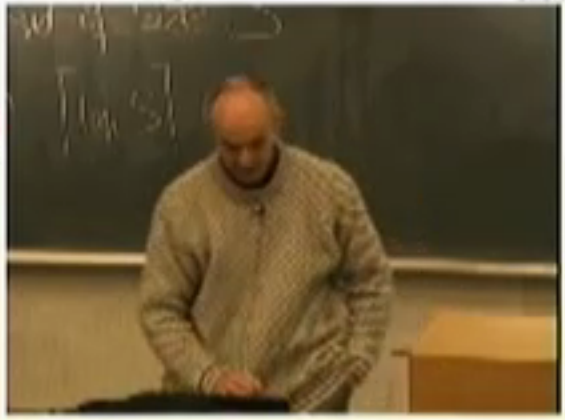
Information learnt: 0.022720

Latest outcome: n



P(n): 63/64 = 0.9843	h(n): 0.0227	hTOT: 0.02272

```
T=1000;
for t=1:T
  ans=input('-');
  lottery
endfor
20:lewis:/home/mackay/itp/octave/lottery>
20:lewis:/home/mackay/itp/octave/lottery>
20:lewis:/home/mackay/itp/octave/lottery>
```



$$P(r_1 = n) = \frac{63}{64}$$

$$h(r_1 = n) = \log_2 \frac{64}{63}$$

$$= 0.0227$$

Submarine

Restart

Submarine

Dismiss

Unhit subs: 1

Unhit squares: 62

Probabilities:

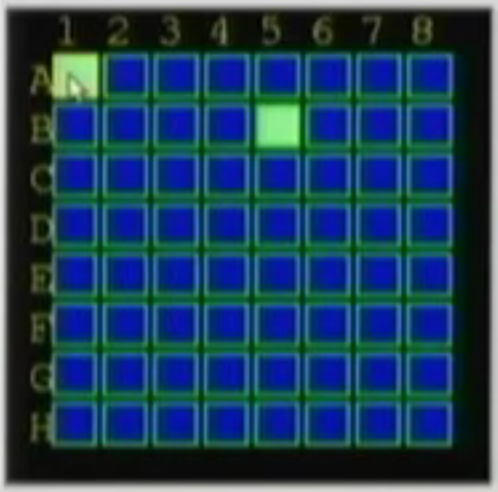
P(y): 0/1 0.0

P(n): 1/1 1.0

Entropy: 0

Information learnt: 0.045803

Latest outcome: n



P(n): 63/64 = 0.9843	h(n): 0.0227	hTOT: 0.02272
P(n): 62/63 = 0.9841	h(n): 0.0230	hTOT: 0.04580

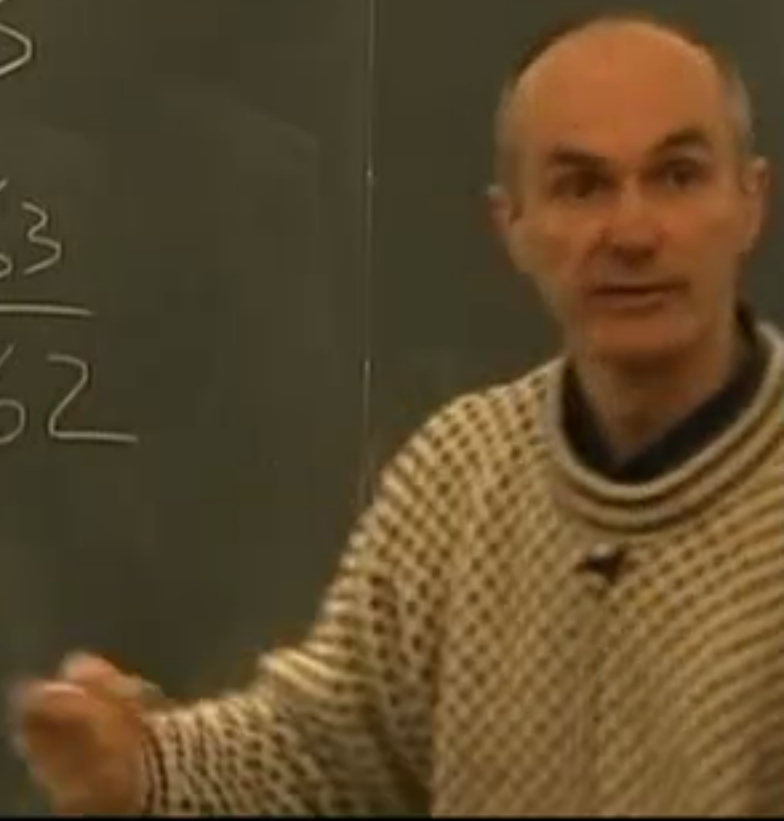
```
T=1000;
for t=1:T
  ans=input('-');
  lottery
endfor
20:lewis:/home/mackay/itp/octave/lottery>
20:lewis:/home/mackay/itp/octave/lottery>
20:lewis:/home/mackay/itp/octave/lottery>
```



$$P(r_2 = n) = \frac{62}{63}$$

$$h = \log_2 \frac{63}{62}$$

$$= 0.023$$



$$h_{TOT} = \log_2 \frac{64}{63} + \log_2 \frac{63}{62}$$

Submarine

Restart

Submarine

Dismiss

Unhit subs: 1

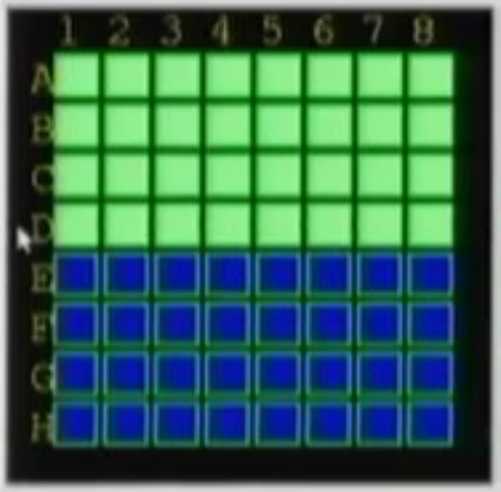
Unhit squares: 32

P(y):

P(n):

Probabilities:

Entropy:

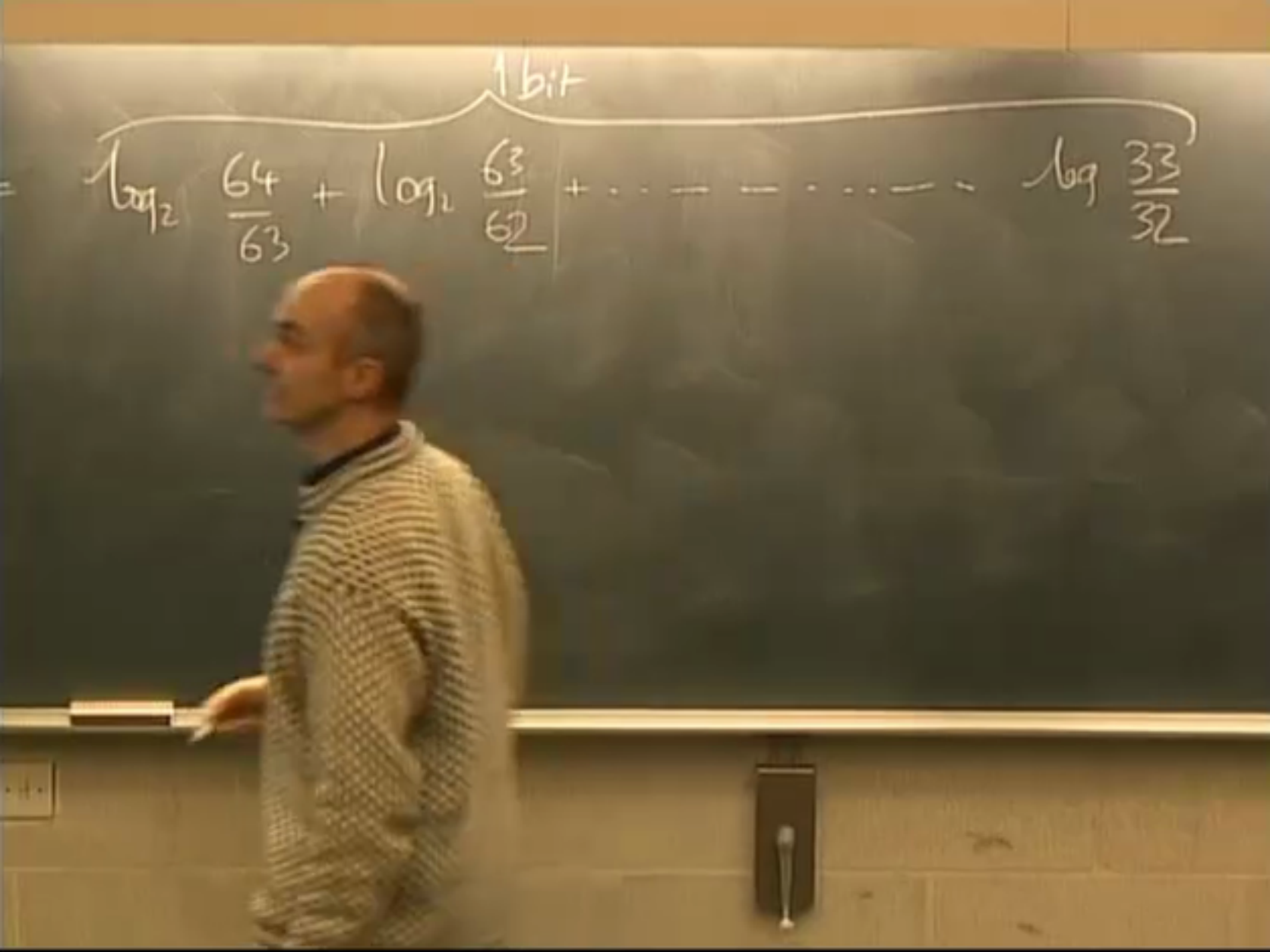


Information learnt: 1.000000

Latest outcome: n

P(n): 40/41 = 0.9756	h(n): 0.0356	hTOT: 0.67807
P(n): 39/40 = 0.975	h(n): 0.0365	hTOT: 0.71459
P(n): 38/39 = 0.9743	h(n): 0.0374	hTOT: 0.75207
P(n): 37/38 = 0.9736	h(n): 0.0384	hTOT: 0.79054
P(n): 36/37 = 0.9729	h(n): 0.0395	hTOT: 0.83007
P(n): 35/36 = 0.9722	h(n): 0.0406	hTOT: 0.87071
P(n): 34/35 = 0.9714	h(n): 0.0418	hTOT: 0.91253
P(n): 33/34 = 0.9705	h(n): 0.0430	hTOT: 0.95560
P(n): 32/33 = 0.9696	h(n): 0.0443	hTOT: 1.00000

```
T=1000;
for t=1:T
  ans=input('-');
  lottery
endfor
20:lewis:/home/mackay/itp/octave/lottery>
20:lewis:/home/mackay/itp/octave/lottery>
20:lewis:/home/mackay/itp/octave/lottery>
```


1 bit
$\log_2 \frac{64}{63} + \log_2 \frac{63}{62} + \cdots + \log \frac{33}{32}$

Submarine

Restart

Submarine

Dismiss

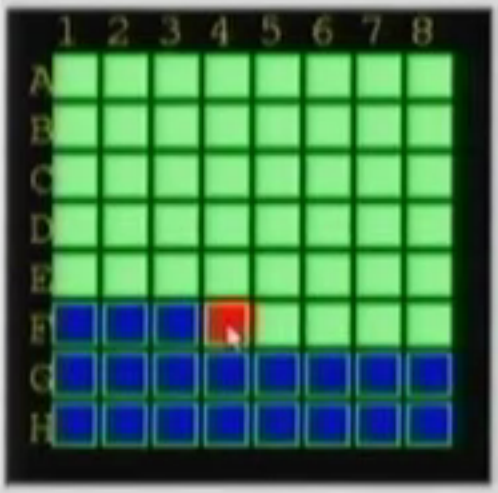
Unhit subs: 0

Unhit squares: 19

Probabilities:

P(y):	P(n):
1/1	19/20
1.0	0.0

Entropy: 0



Information learnt: 6.000000

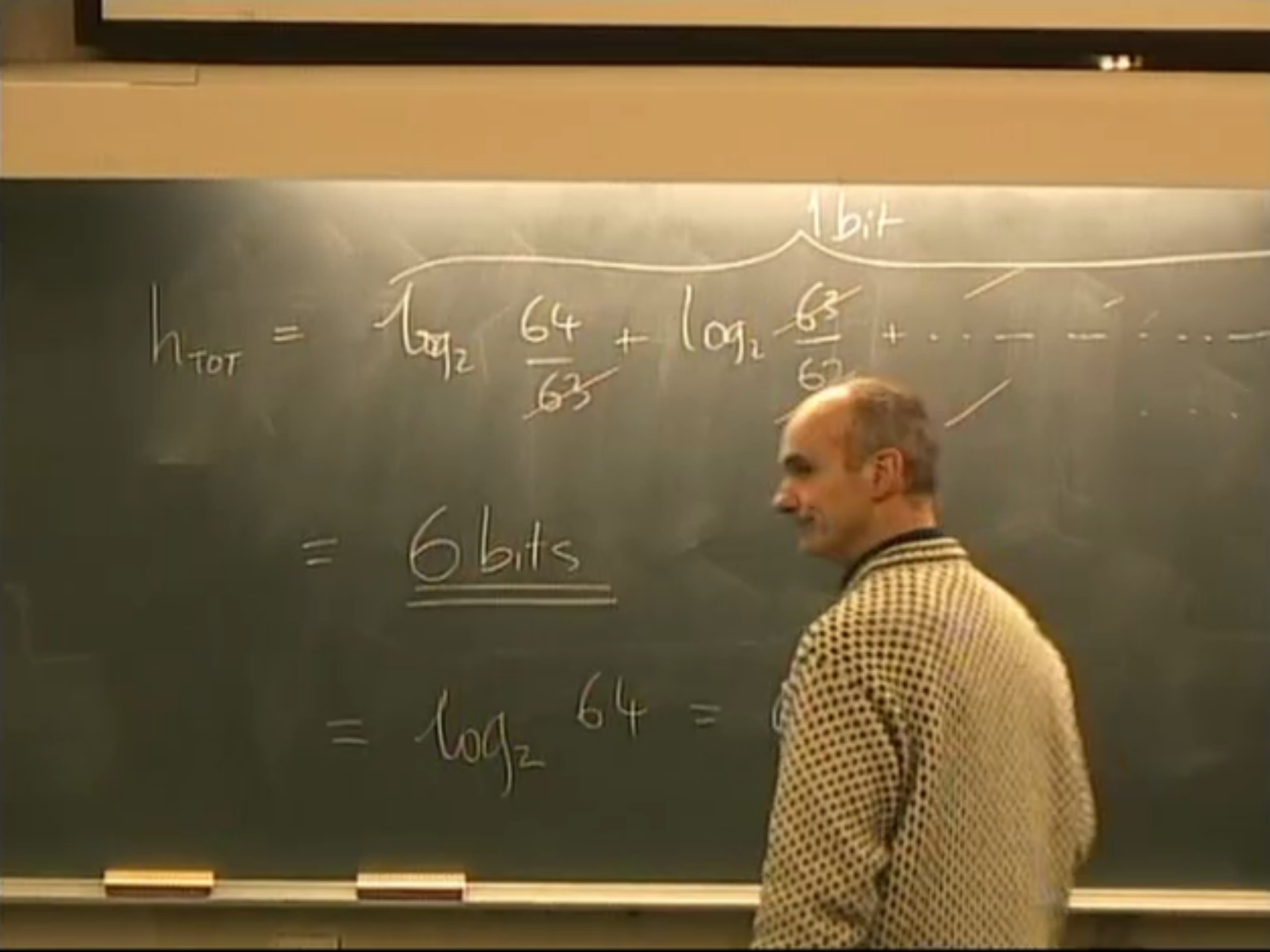
Latest outcome: y

P(n): 27/28 = 0.9642	h(n): 0.0524	hTOT: 1.24511
P(n): 26/27 = 0.9629	h(n): 0.0544	hTOT: 1.29956
P(n): 25/26 = 0.9615	h(n): 0.0565	hTOT: 1.35614
P(n): 24/25 = 0.96	h(n): 0.0588	hTOT: 1.41503
P(n): 23/24 = 0.9583	h(n): 0.0614	hTOT: 1.47643
P(n): 22/23 = 0.9565	h(n): 0.0641	hTOT: 1.54056
P(n): 21/22 = 0.9545	h(n): 0.0671	hTOT: 1.60768
P(n): 20/21 = 0.9523	h(n): 0.0703	hTOT: 1.67807
P(y): 1/20 = 0.05	h(y): 4.3219	hTOT: 6.00000

```
T=1000;
for t=1:T
  ans=input('-');
  lottery
endfor
20:lewis:/home/mackay/itp/octave/lottery>
20:lewis:/home/mackay/itp/octave/lottery>
20:lewis:/home/mackay/itp/octave/lottery>
```

$$\log \frac{33}{32} + \log \frac{32}{31} + \cdots + \log \frac{21}{20} + \log_2 20$$

4.3 bits


1 bit
$h_{TOT} = \log_2 \frac{64}{63} + \log_2 \frac{63}{62} + \ldots$
$= 6$ bits
$= \log_2 64 =$

The Bent Coin Lottery

A coin with $p_1 = 0.1$ will be tossed $N = 1000$ times.

The outcome is $\mathbf{x} = x_1 x_2 \ldots x_N$.

e.g., $\mathbf{x} = 000001001000100\ldots00010$

You can buy any of the 2^N possible tickets for £1 each, before the coin-tossing.

If you own ticket $\mathbf{x}$, you win £1,000,000,000.

Lottery tickets available

2^N

0000000000.....00000
0000000000.....00001
0000000000.....00010
0000000000.....00011
0000000000.....00100
0000000000.....00101
0000000000.....00110
0000000000.....00111
⋮
0010000001.....01000
⋮
1111111111.....11101
1111111111.....11110
1111111111.....11111

Q If you are forced to buy *one* ticket, which would you buy?

Q To have a 99% chance of winning, at lowest possible cost, which tickets would you buy?

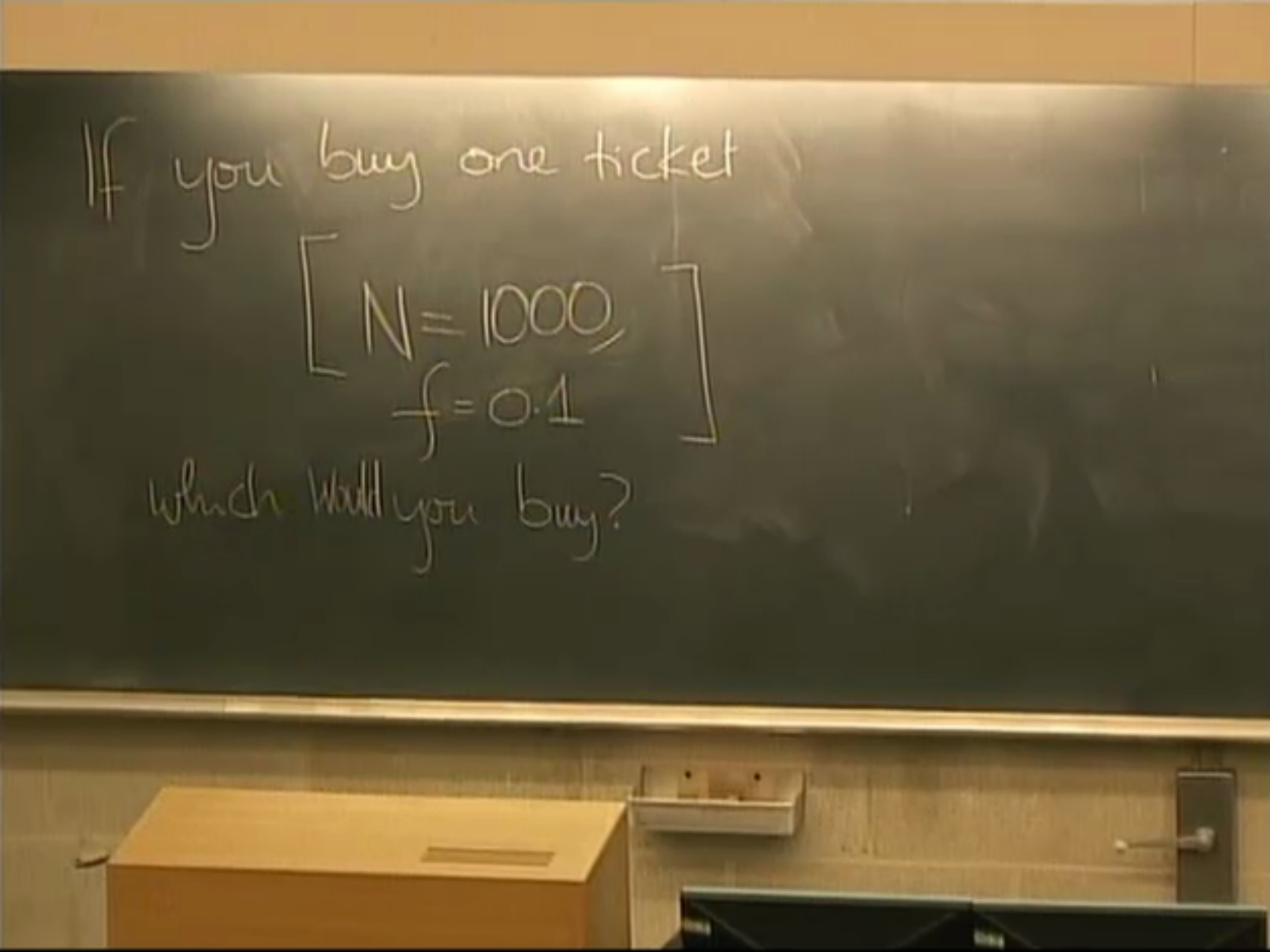
- And how many tickets is that?

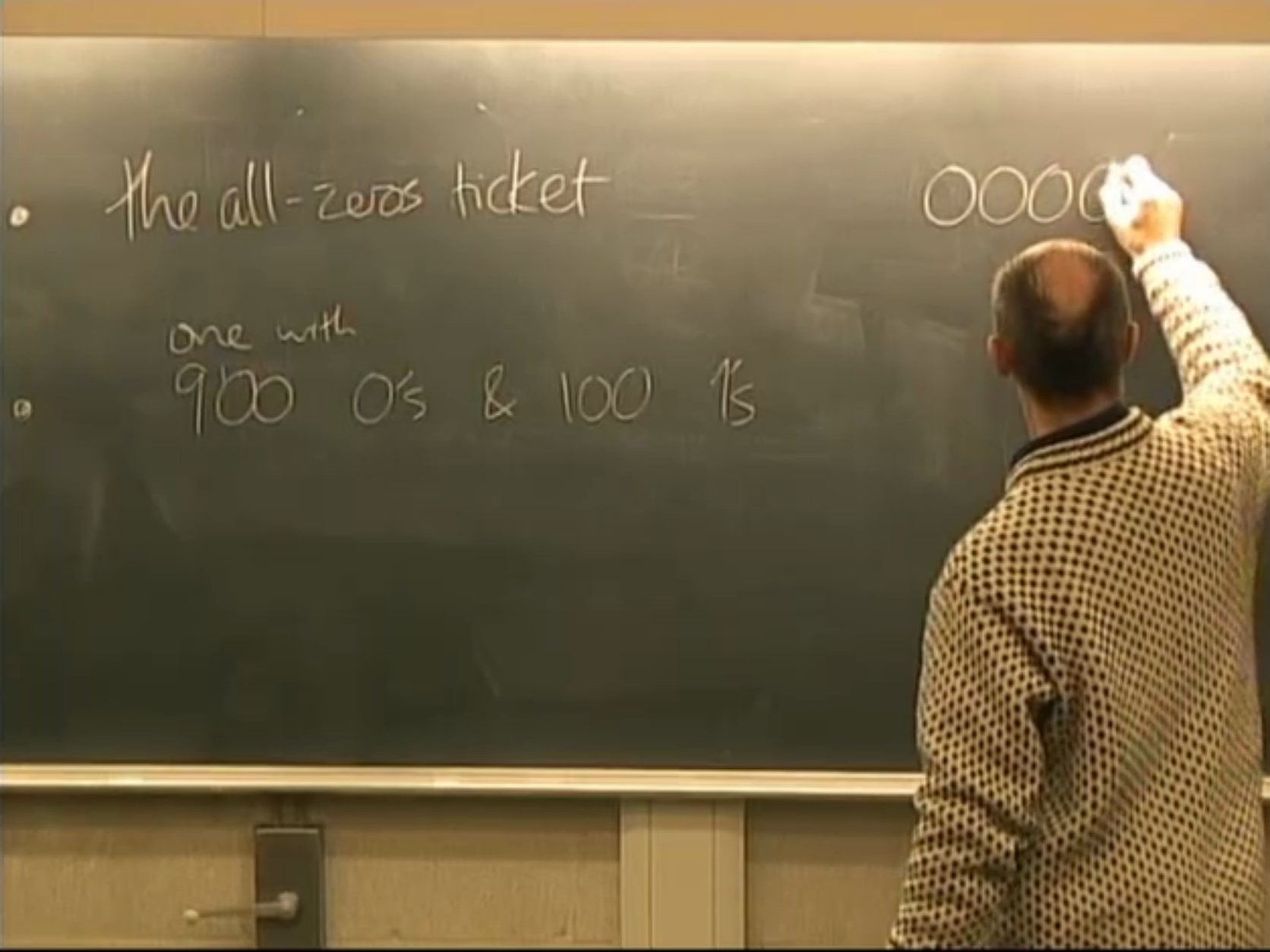
  Express your answer in the form $2^{(\cdots)}$.

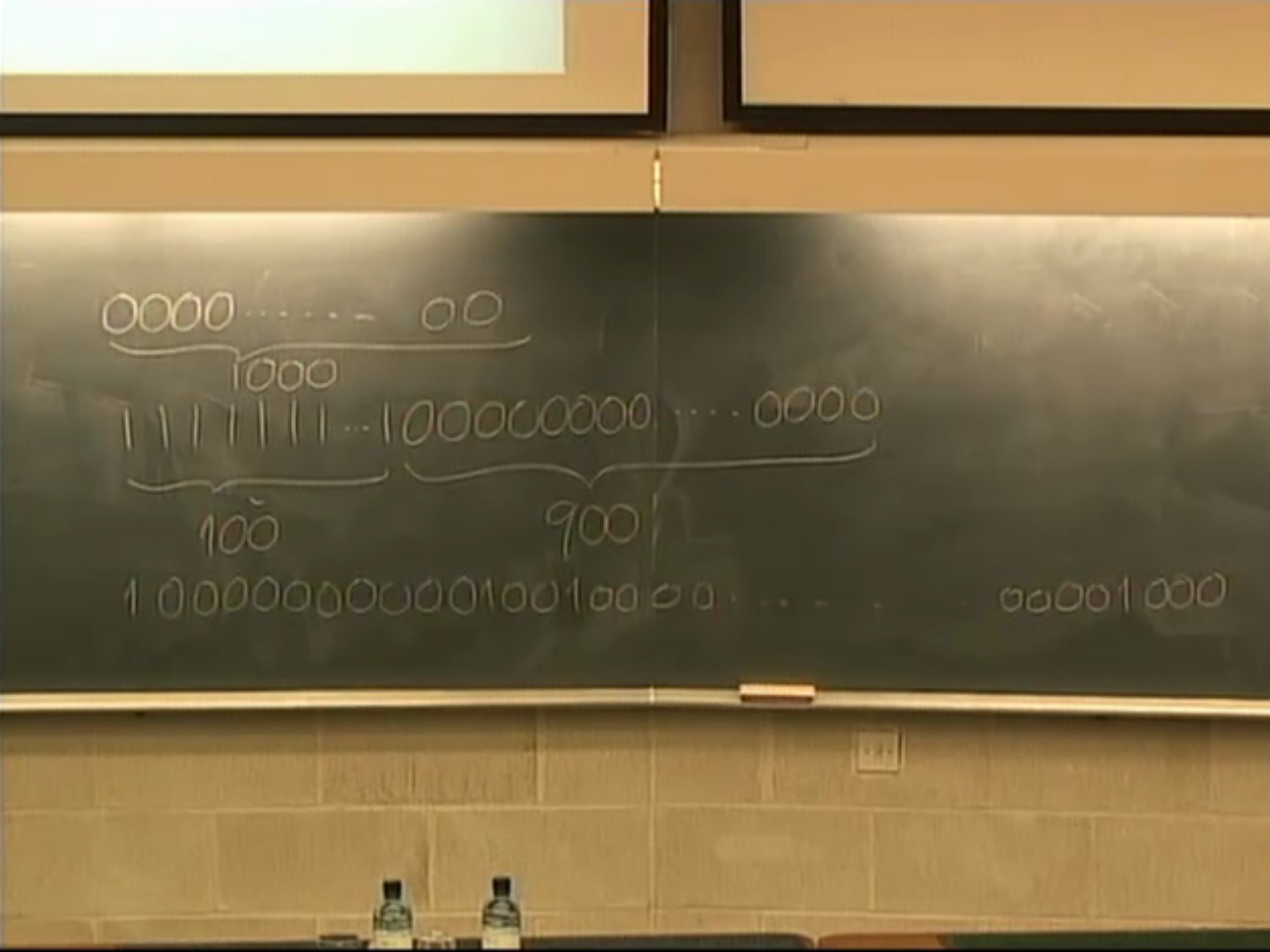
If you buy one ticket

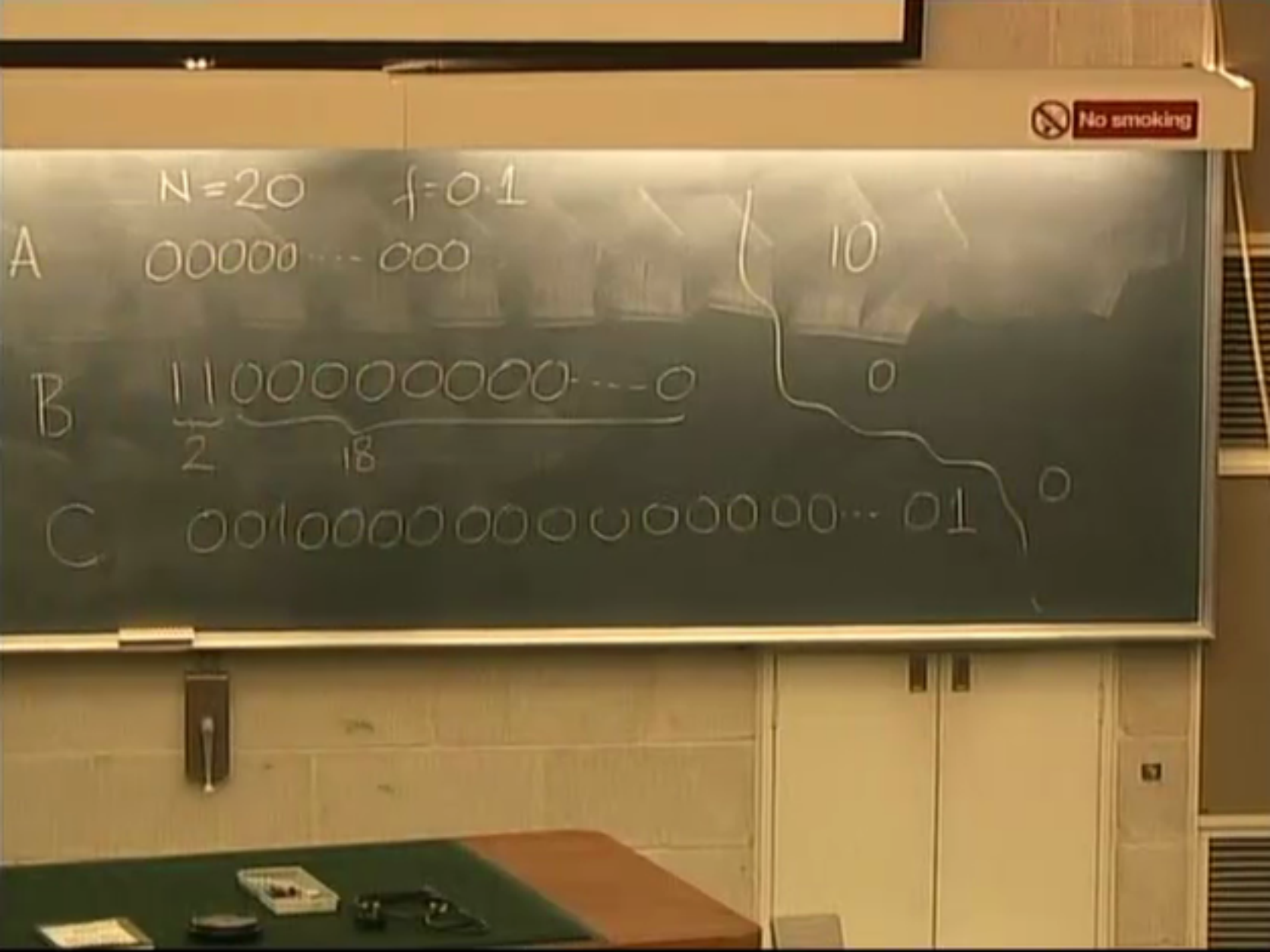
$$\left[\begin{array}{l} N = 1000 \\ f = 0.1 \end{array}\right]$$

Which would you buy?




the all-zeros ticket
0000
one with
900 0's & 100 1's


0000 00
1000
1111111..1
00000000 0000
100
900
1000000000010010000 00001000

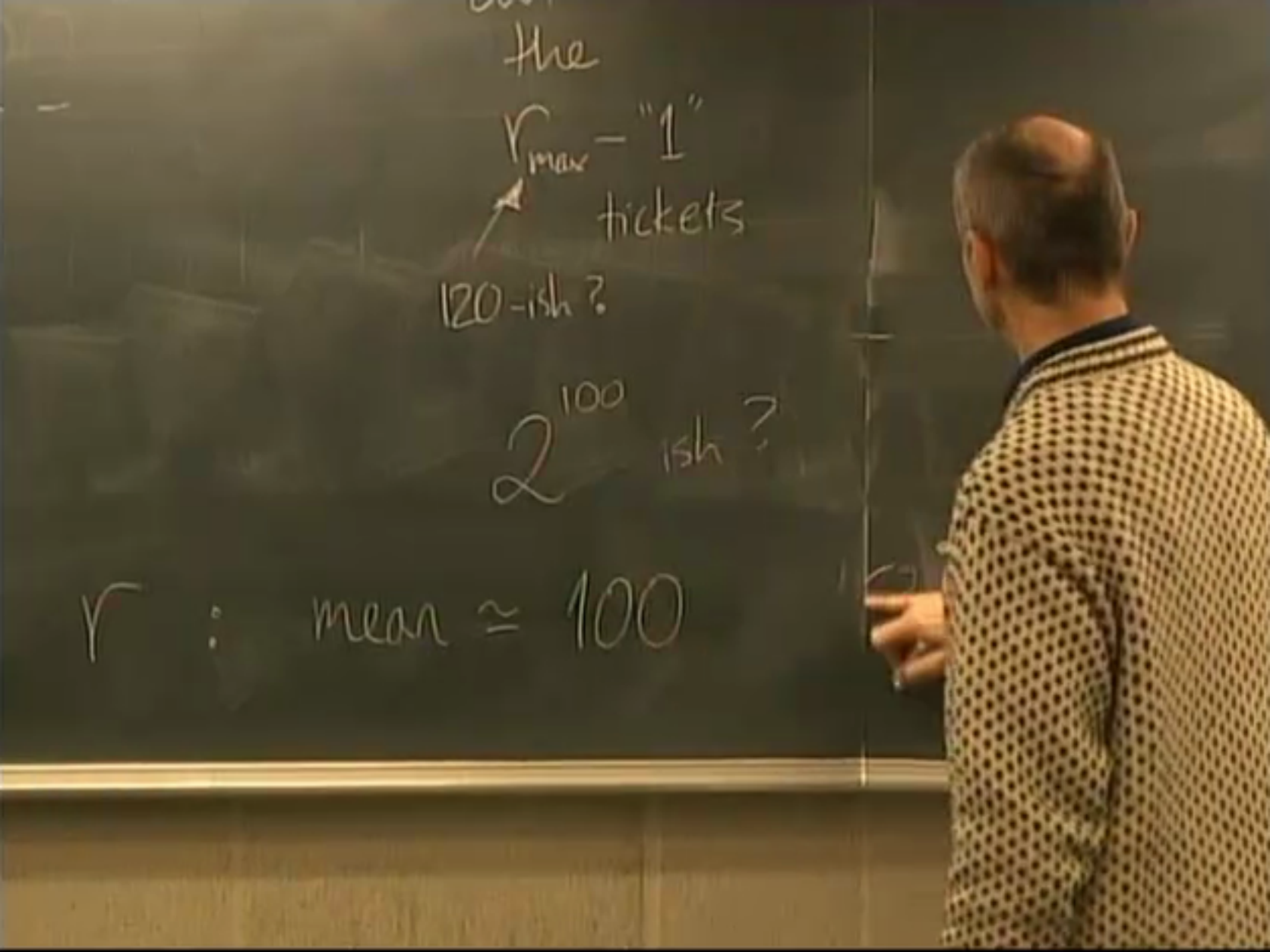

No smoking
N=20 f=01
A 00000···000
10
B 11 00000000000···0
2 18
0
C 001000000000000000···01
0

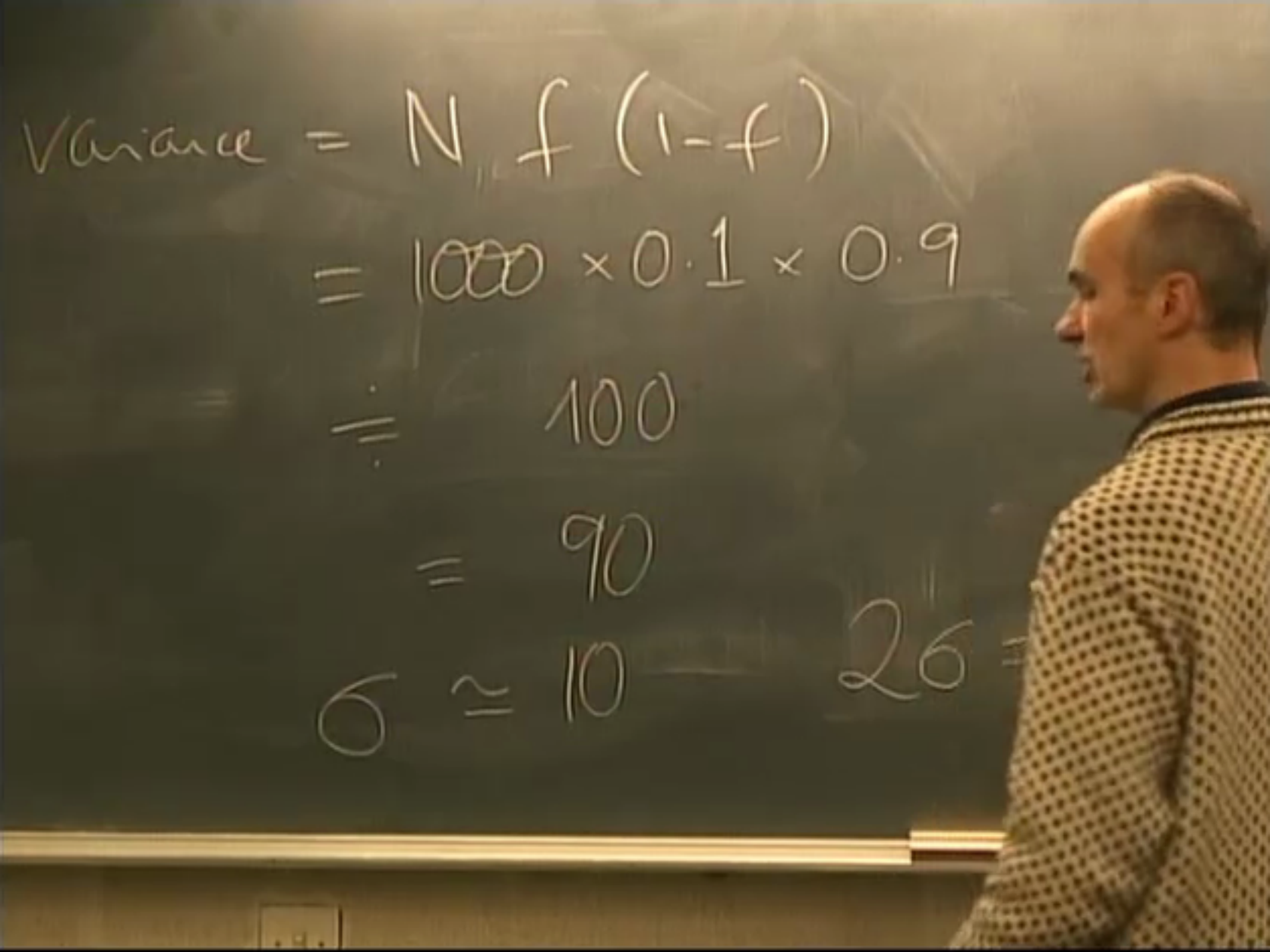
To have a 99% chance

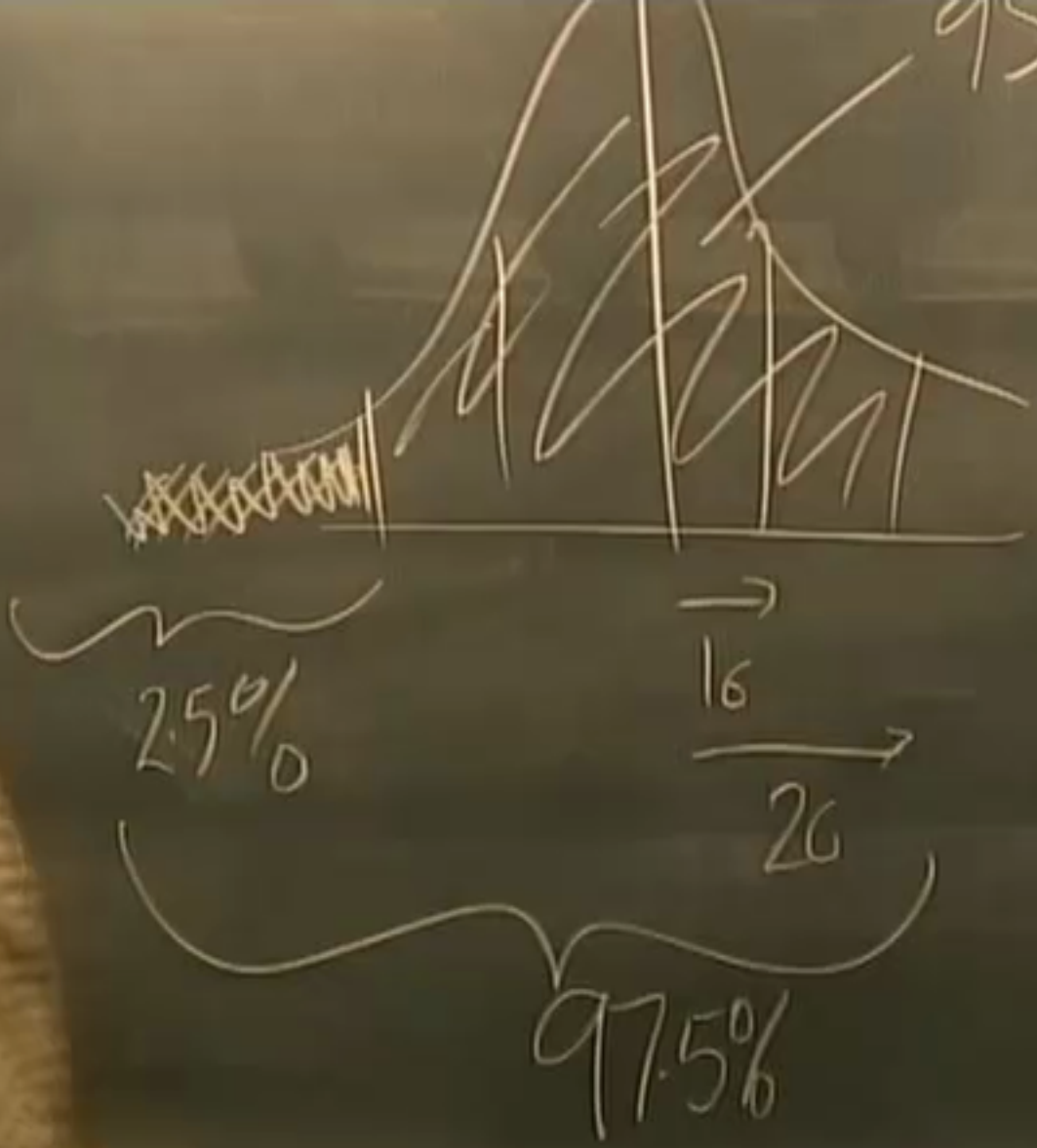
$$\left[\begin{array}{c} N = 1000 \\ f = 0.1 \end{array}\right]$$

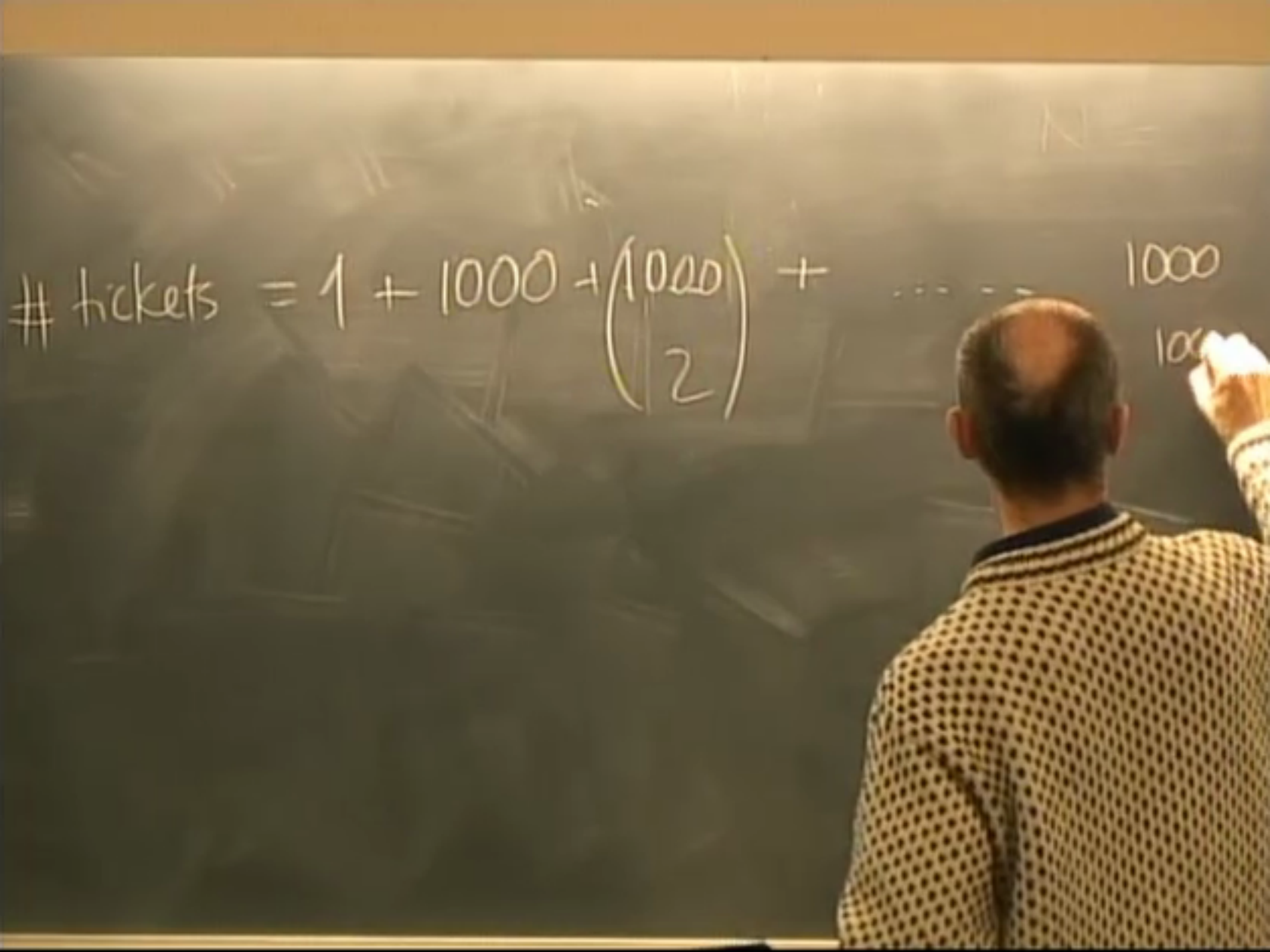
which would you buy?

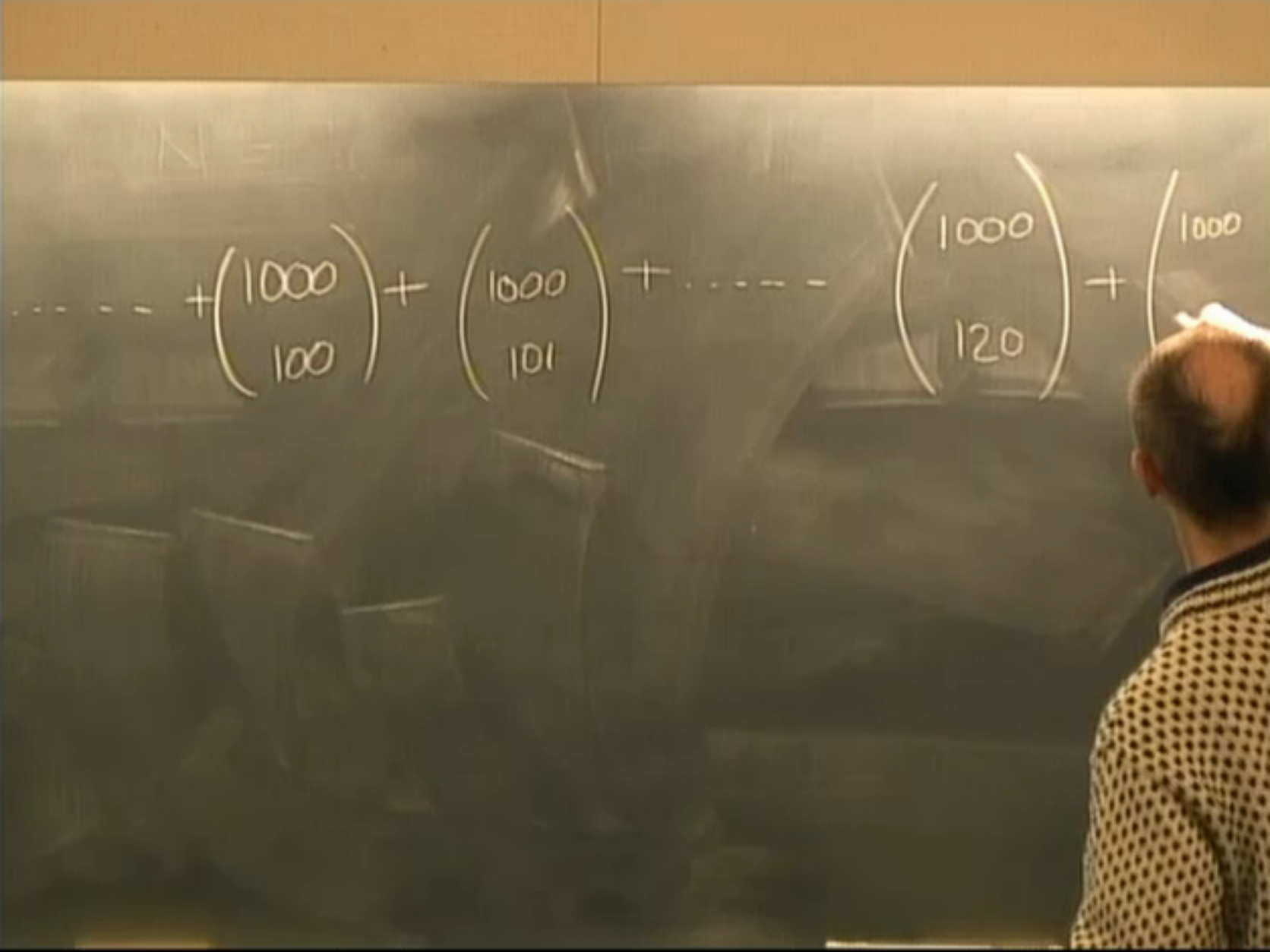
00…00, all the 1-"1" tickets, all the 2-"1" tix,


the
r_{max} – "1"
tickets
120-ish?
2^{100} ish?
r : mean ≃ 100


Variance = N f (1-f)
= 1000 × 0.1 × 0.9
≒ 100
= 90
σ ≃ 10
26


95%
2.5%
1σ
2σ
97.5%

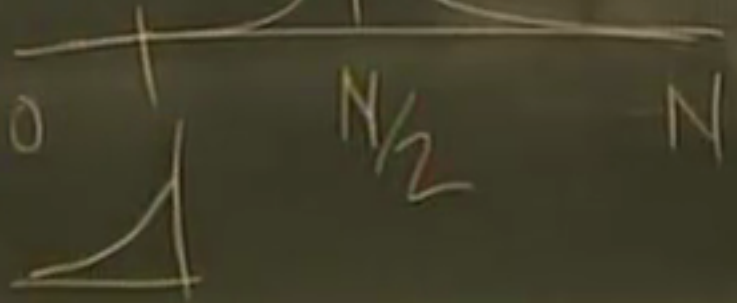

#tickets = 1 + 1000 + $\binom{1000}{2}$ + ... 1000


. . . . . + $\binom{1000}{100}$ + $\binom{1000}{101}$ + ... $\binom{1000}{120}$ + $\binom{1000}{$

No smoking

$$\binom{1000}{120} + \binom{1000}{121} + \binom{1000}{122} + \binom{1000}{123}.$$

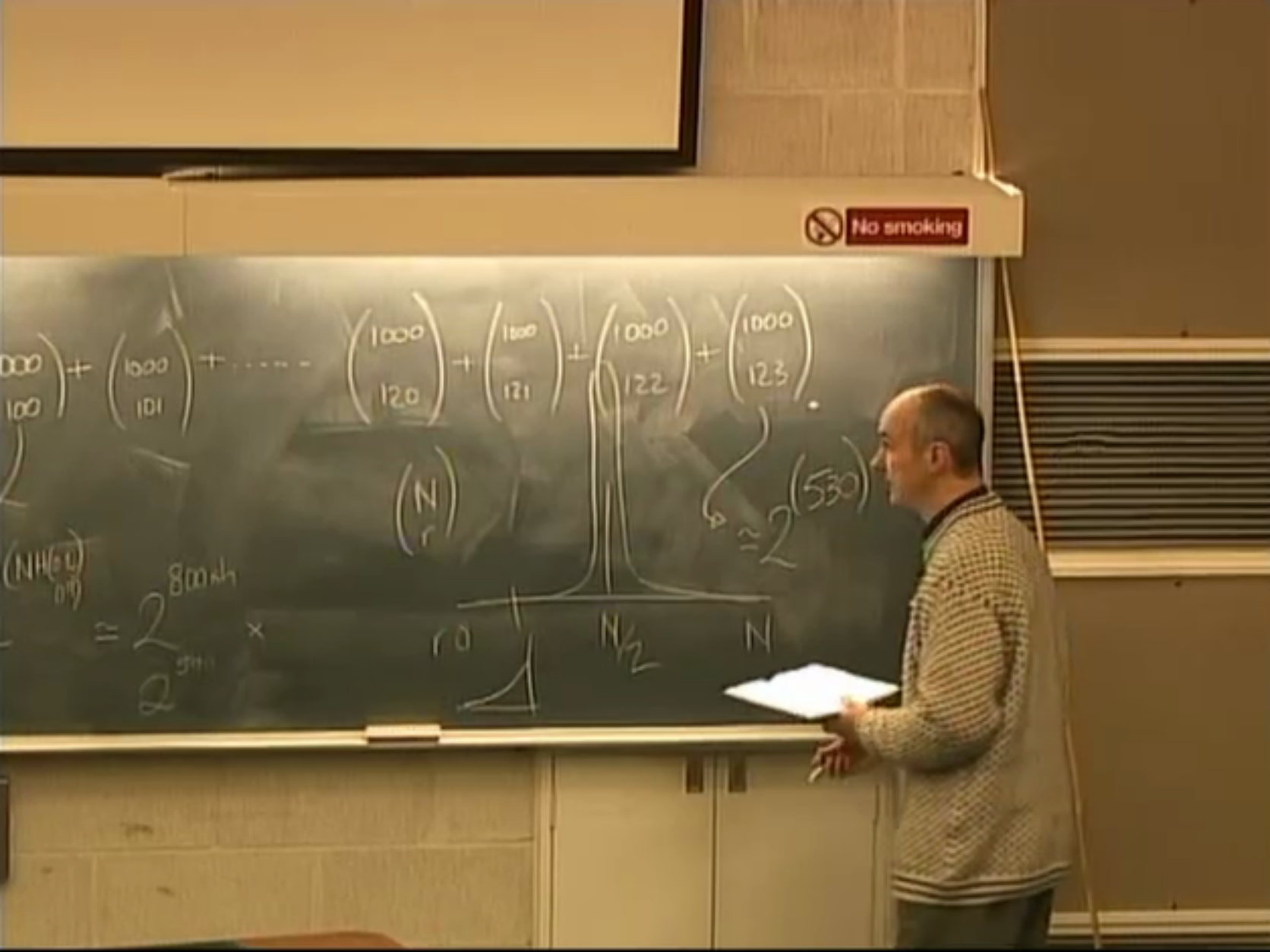
$$\binom{N}{r}$$



$$\#\,\text{tickets} = 1 + 1000 + \binom{1000}{2} + \cdots + \binom{1000}{100} + \binom{1000}{\cdots}$$

$$\log \frac{N!}{(N-r)!\,r!} \simeq N H_2\left(\frac{r}{N}\right)$$

$$\simeq 2^{N H(0.1)} \simeq 2^{N H(0.9)} \simeq$$

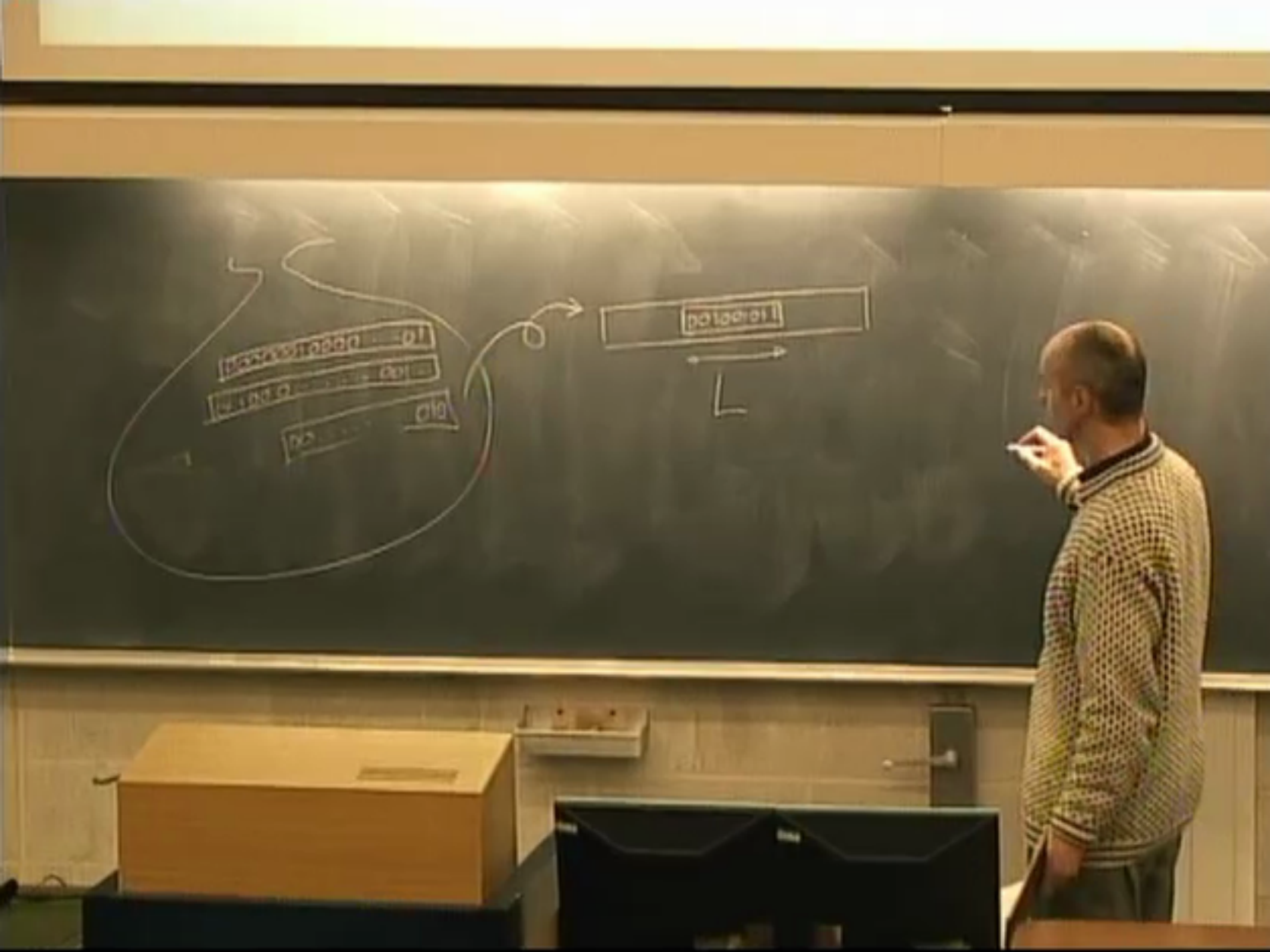

No smoking
$\binom{1000}{100} + \binom{1000}{101} + \cdots + \binom{1000}{120} + \binom{1000}{121} + \binom{1000}{122} + \binom{1000}{123}$
$\binom{N}{r}$
$\simeq 2^{(530)}$
$\simeq 2^{800 H_{\frac{1}{4}}}$
r 0
N/2
N

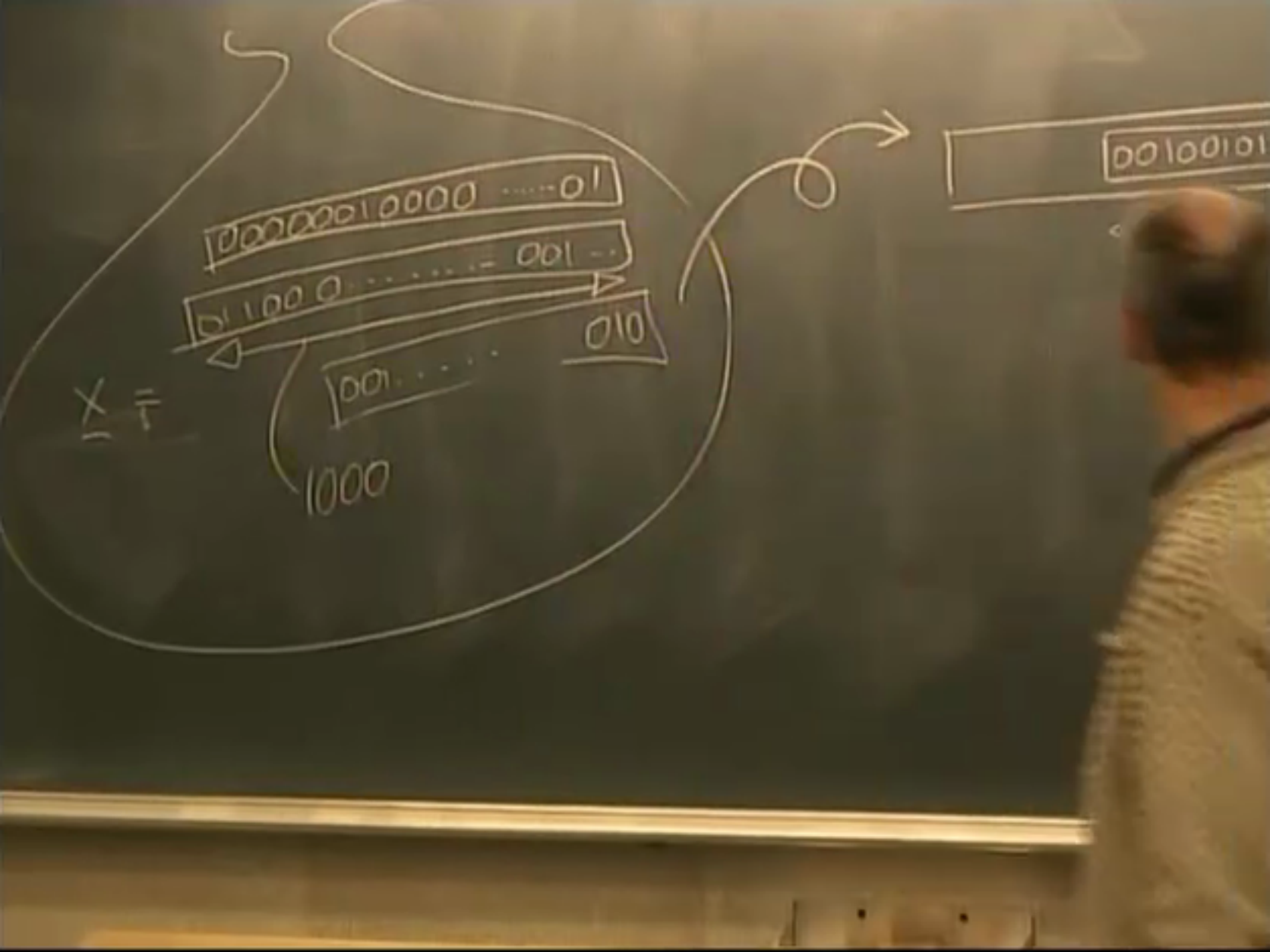
$$\left(N \right)$$

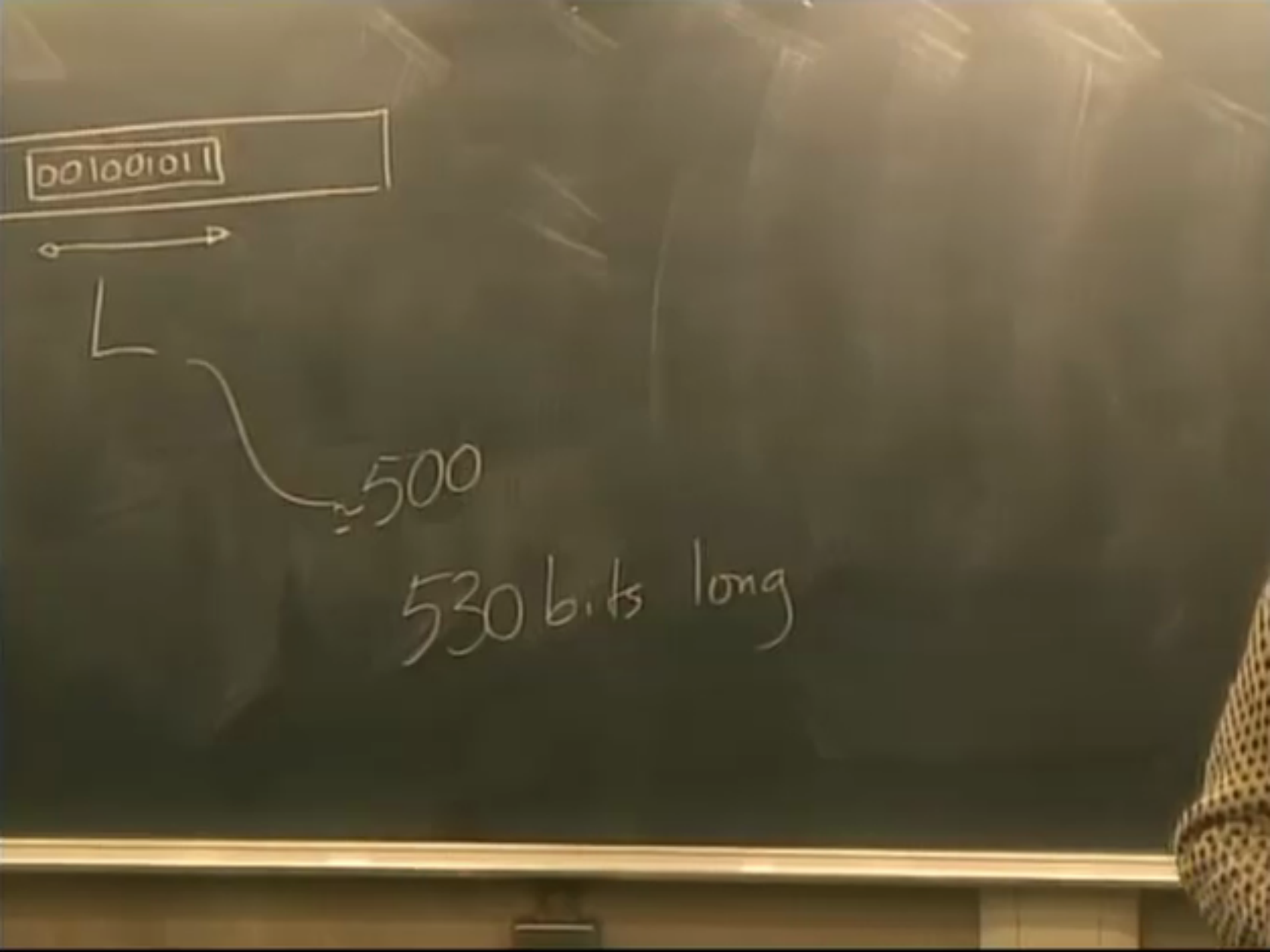
$$fN \pm 2.3\sqrt{f(1-f)N}$$

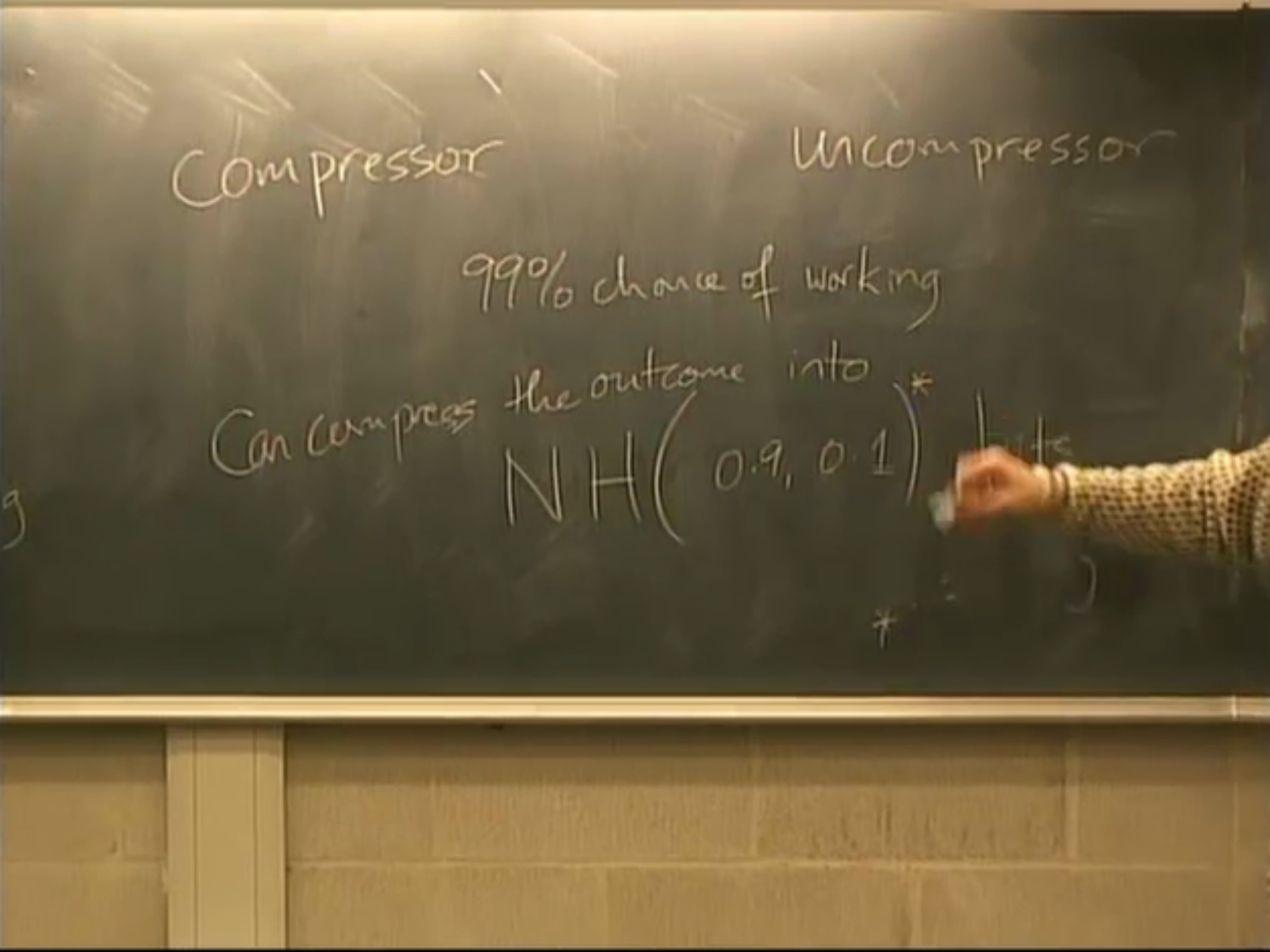
$NH_2(f)$ $(+\sqrt{N}\ldots)$

$= 2$




1000


00100101 1
L
500
530 bits long


Compressor
Uncompressor
99% chance of working
Can compress the outcome into
NH(0.9, 0.1)*
bits

Source coding theorem

3. Source coding theorem –

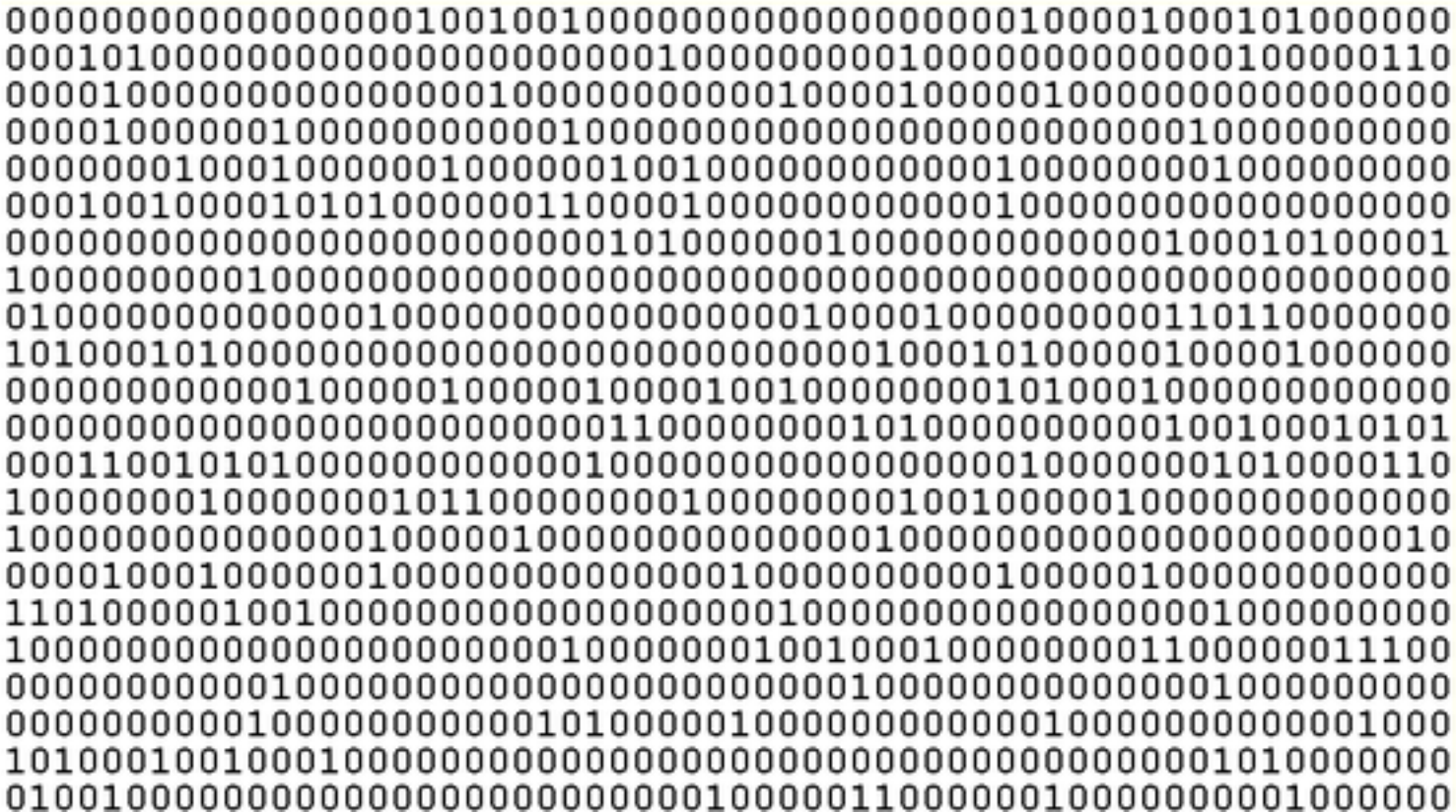
   N outcomes from a source X can be compressed into roughly $NH(X)$ bits.

Proved by counting the typical set

Read chapter 4 to see the full proof and the general definition of typicality

How to compress a redundant file,

practically?



Project:

Invent a compressor and uncompressor for a source file of $N = 10,000$ bits, each having probability $f = 0.01$ of being a 1.
Implement them and/or
estimate how well your method works.

- Also: exercises 5.22, 5.26, 5.27
- Reading: Chapters 1-6